

**Adjective Ordering in Referential Communication:  
Subjectivity, Discriminatory Strength, and Incremental Pragmatics**

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### Abstract

Adjective ordering preferences are robust across languages, and recent work suggests that they reflect both stable lexical-semantic asymmetries and context-sensitive pressures on efficient reference resolution. The present paper develops a Rational Speech Act model that brings these traditions together by crossing two dimensions in a  $2 \times 2$  comparison: speaker architecture (global vs. incremental) and semantic regime (context-fixed vs. context-updating). Under context-updating semantics, the interpretation of a gradable adjective is recomputed from the listener’s running posterior during composition, making adjective order non-commutative; under context-fixed semantics, interpretations are fixed from the scene. Two referential experiments in German provide the empirical target: a slider-rating study that measures graded preference for competing adjective orders and a free-production study that measures actual utterance choice. Across both experiments, ordering preferences vary systematically with the discriminatory value of the adjectives in the current scene, with speakers tending to place the more discriminating adjective earlier in the string, while still preserving a baseline size-first tendency. Bayesian model comparison shows that incremental production is the dominant mechanism across both datasets. In the production data, context-updating semantics provides a significant further improvement, but only when paired with an incremental speaker — an interaction indicating that the two mechanisms are synergistic. Simulation results confirm that a substantial baseline ordering effect already follows from context-sensitive restrictive composition alone. We argue that adjective ordering in referential communication is best understood as a graded adaptation to efficient, incremental reference resolution: subjectivity provides a baseline bias, discriminatory strength modulates that bias in context, and incremental production with context-updating semantics best captures how speakers realize these pressures online.

*Keywords:* adjective ordering, Rational Speech Act, incremental pragmatics, communicative efficiency, subjectivity

## Adjective Ordering in Referential Communication: Subjectivity, Discriminatory Strength, and Incremental Pragmatics

### 1 Introduction

When multiple adjectives modify a noun, speakers across typologically diverse languages prefer similar orderings — “big brown bag” over “brown big bag” (Scontras, 2023; Sproat & Shih, 1991a). Previous research (for an overview, see Scontras, 2023) has linked these preferences to meaning-based factors such as closeness to the noun and inherentness (Martin, 1969; Sweet, 1898; Whorf, 1945), to class-based hierarchies (Cinque, 1994; Dixon, 1982; Quirk, Greenbaum, Leech, & Svartvik, 1972; Scott, 2002), and to information-theoretic measures (Dyer, Torres, Scontras, & Futrell, 2023; Futrell, Dyer, & Scontras, 2020; Hahn, Degen, Goodman, Jurafsky, & Futrell, 2018). However, no single factor has proven sufficient; even the best individual predictors leave complementary variance unexplained (Dyer et al., 2023). Furthermore, the preferences are context-sensitive in ways that fixed hierarchies do not fully capture (Fukumura, 2018; Trotzke & Wittenberg, 2019).

We focus on two lines of recent work that ground ordering preferences in communicative efficiency. Scontras, Degen, and Goodman (2017) showed that *subjectivity* is a strong predictor of preferred adjective distance from the noun, and subsequent work argued that this regularity follows from restrictive composition and sequential context update: less subjective adjectives contribute more useful information at the inner, more restricted stage of modification (Franke, Scontras, & Simonic, 2019; Scontras, Bar-Sever, Kachakeche, Rosales Jr, & Samonte, 2020; Scontras, Degen, & Goodman, 2019). Separately, Fukumura (2018) demonstrated that speakers in referential tasks tend to place the more *discriminating* adjective first, maximising informativeness in the given context, and Rubio-Fernández, Mollica, and Jara-Ettinger (2021) generalised this into an *incremental efficiency* hypothesis: the utility of a modifier should be evaluated at the point in the utterance where it is encountered, not only after the full description has been heard.

Both lines of work appeal to communicative efficiency, but they locate it at different

levels — one in the compositional semantics, the other in the online production process. The central question of this paper is how these two levels interact. Subjectivity provides a stable ordering bias, but the communicative context determines how strongly that bias is expressed in a particular scene. When one property suffices for identification and another does not, the more useful property may exert pressure for earlier mention even when it conflicts with the canonical order. What is missing is a unified account that separates the contribution of stable compositional asymmetries from context-sensitive, online production pressures and tests whether these mechanisms are independently additive or synergistic.

We address three questions. First, can canonical ordering preferences already arise from *context-sensitive restrictive composition* without an incremental production mechanism? Second, does an explicitly *incremental* speaker — one that chooses modifiers word by word — add explanatory value beyond a global, utterance-level evaluation? Third, does *context-updating semantics*, in which the meaning of a gradable adjective is recomputed from the listener’s running posterior, provide a further benefit — and is that benefit contingent on incremental production?

To answer these questions, we develop a Rational Speech Act (RSA) model (Degen, 2023; Goodman & Frank, 2016) that crosses two dimensions in a  $2 \times 2$  comparison: *speaker architecture* (global vs. incremental) and *semantic regime* (context-fixed vs. context-updating). The global speaker evaluates complete utterances; the incremental speaker chooses one modifier at a time. Under context-updating semantics, the interpretation of “big” depends on what the listener already believes after hearing earlier words, so that “big blue” and “blue big” need not be equivalent; under context-fixed semantics, the size threshold is determined once from the scene. All four model variants are fitted to data from two complementary referential experiments in German — a slider-rating study measuring graded ordering preferences and a constrained production study measuring actual utterance choice — in a shared paradigm that varies discriminatory strength across scenes. Model comparison uses Bayesian leave-one-out cross-validation (PSIS-LOO; Vehtari,

Gelman, & Gabry, 2017), which quantifies how much predictive accuracy each mechanism adds and detects their interaction. A simulation study complements the empirical work by crossing both speaker architectures with both semantic regimes to identify which ordering effects require context-sensitive composition, which require incremental production, and which emerge only from their interaction.

Both experiments confirm that ordering preferences vary systematically with discriminatory strength: speakers tend to place the more informative adjective earlier while preserving a baseline size-first tendency. The simulation shows that recursive semantics is necessary for the baseline size-first advantage: the global speaker produces the canonical ordering only under context-updating semantics, where threshold adaptation makes the two orderings informationally non-equivalent. In the model comparison, incremental production emerges as the dominant mechanism across both datasets. In the production data, context-updating semantics provides a significant further improvement, but only when paired with an incremental speaker — an interaction indicating that the two mechanisms are synergistic. Figure 1 provides a schematic overview of the modelling framework, experiments, and key findings. The remainder of the paper is structured as follows. Section 2 reviews the three lines of prior work that the present account integrates — subjectivity-based, efficiency-based, and incremental-pragmatic. Section 3 introduces the modelling framework conceptually, before the experiments, with the formal specification deferred to Section 5. Section 4 reports the two referential studies: a slider-rating task (Section 4.2) and a free-production task (Section 4.3). Section 6 fits the four RSA variants to both datasets and quantifies the contribution of each mechanism using PSIS-LOO. Section 7 crosses the same  $2 \times 2$  in a parameter sweep to identify which ordering effects already follow from compositional restriction alone. Section 8 draws the three threads together.

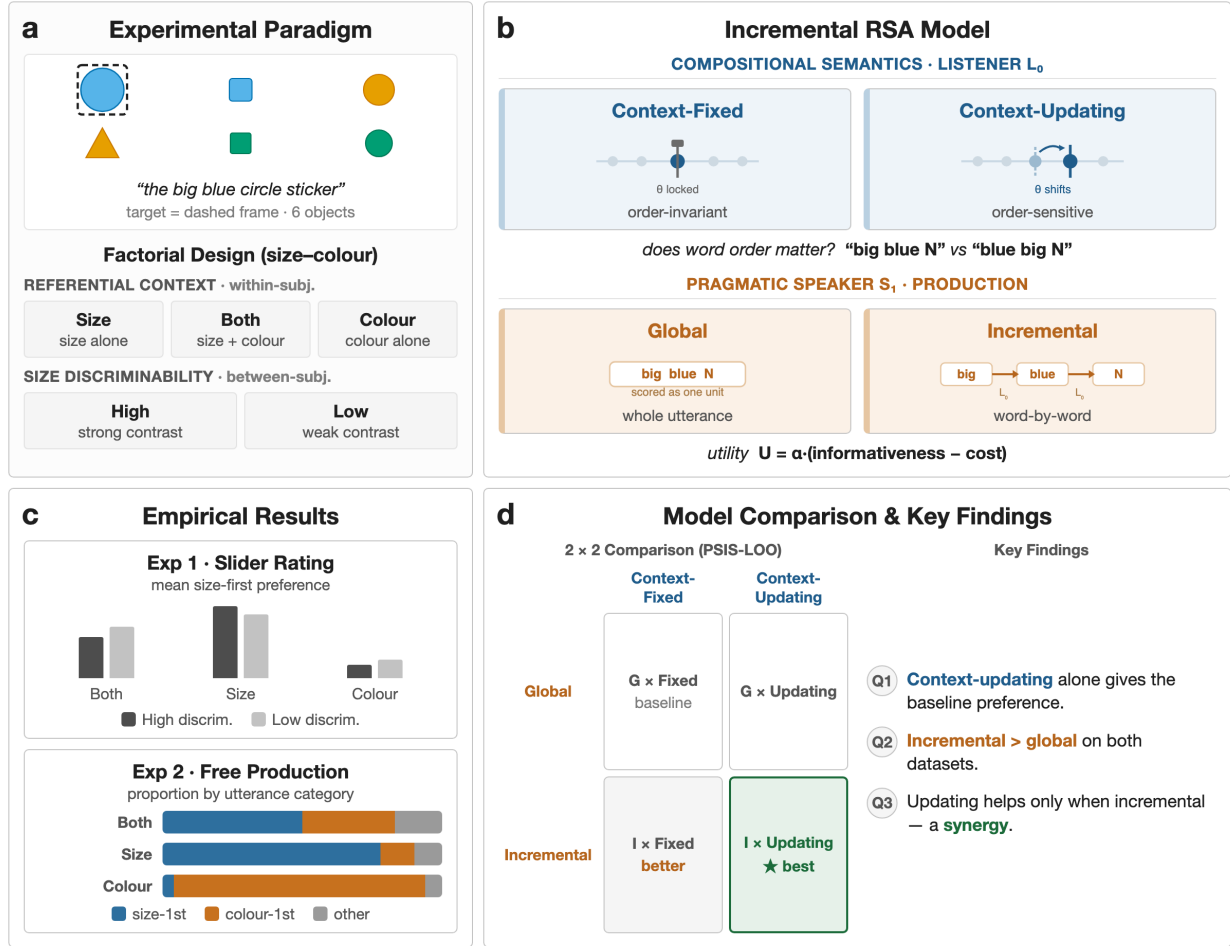


Figure 1

Overview of the present study. (a) Experimental paradigm: six-object displays varying in referential context (within-subj.) and size discriminability (between-subj.). (b) The RSA model: compositional semantics (context-fixed vs. context-updating  $\theta$ , right-to-left) crossed with speaker architecture (global vs. incremental, left-to-right). (c) Slider ratings of ordering preference and production proportions by utterance category. (d) 2 × 2 model comparison (PSIS-LOO).

## 2 Background

Three lines of research bear on the question of how speakers choose and order adjectives, yet to date they have developed largely independently: subjectivity-based accounts trace canonical ordering preferences to compositional interpretation,

efficiency-based accounts link adjective choice and order to the discriminatory strength covered by the referential context, and incremental pragmatic models formalize the word-by-word process through which speakers produce utterances. The model developed in this paper integrates all three.

## 2.1 Subjectivity and Adjective Ordering

A central empirical generalization in the adjective-ordering literature is the relation between an adjective’s *subjectivity* and its preferred distance from the noun. Scontras et al. (2017) operationalized subjectivity as susceptibility to faultless disagreement and showed that it explains a large share of variance in English ordering preferences, outperforming earlier predictors such as inherentness and intersectivity. Subsequent cross-linguistic work suggests that this relationship is not peculiar to English, but reflects a broader regularity in adjective ordering (Scontras, 2023; Trainin & Shetreet, 2021).

Scontras et al. (2019) argue that the explanatory core of this effect lies in restrictive composition. Adjectives closer to the noun apply over a denotation that has already been narrowed by outer modifiers. In such a restricted comparison class, a less subjective adjective is advantageous because its interpretation is more stable. Franke et al. (2019) make the same intuition explicit in simulation, showing that subjectivity-aligned orders increase average communicative success under sequential composition with noisy meanings. Crucially, this explanation does not require a word-by-word production mechanism; the ordering pressure can emerge from the compositional system itself.

That conclusion is strengthened by Scontras et al. (2020), who argue that what matters is *incremental semantic restriction*. If nominal structure supports sequential narrowing of the comparison class, then subjectivity-based ordering pressures should arise even without an incremental speaker in the usual process-model sense. This identifies a plausible minimal mechanism for canonical ordering: context-sensitive restrictive composition may already suffice to produce a baseline size-first tendency.

## 2.2 Communicative Efficiency and Discriminatory Strength

The subjectivity account, however, addresses only one source of ordering pressure. A separate line of work has shown that adjective use is sensitive to the demands of the immediate communicative context, including whether a modifier helps the listener identify the referent quickly and efficiently (Degen, Hawkins, Graf, Kreiss, & Goodman, 2020; Rubio-Fernández, 2016; Rubio-Fernandez, 2019). This literature treats adjective choice and adjective order as part of a broader problem of reference generation rather than as the expression of a static lexical hierarchy.

Fukumura (2018) provide a direct precedent for the present experiments. In their production studies, speakers were more likely to place the more discriminating adjective first, consistent with a *discriminatory efficiency hypothesis*: order should reflect which property best distinguishes the target in the current scene. This is a different source of pressure from subjectivity, because it is inherently local. The same adjective may be highly useful in one display and much less useful in another.

Rubio-Fernández and colleagues place this idea in a broader framework of *incremental efficiency* (Rubio-Fernandez & Jara-Ettinger, 2020; Rubio-Fernández et al., 2021). Their central claim is that the efficiency of a referential expression should be evaluated as the utterance unfolds, not only after the full description has been heard. From this perspective, word order matters because it determines how the listener’s search space is narrowed over time. Prenominal and postnominal systems therefore create different pragmatic affordances, and even redundant modifiers can be useful if they facilitate early search. What matters is not merely whether a property contributes to unique reference in the end, but whether it has discriminatory value at the relevant stage of processing.

For the present paper, this distinction motivates the central empirical manipulation. Within the modeled size–colour subset, the experiments vary whether size is sufficient, colour is sufficient, or whether both properties are necessary for identification, and they additionally vary how discriminable the size contrast is. These manipulations operationalize the



context-sensitive component of the theory: if adjective ordering is responsive to discriminatory strength, then preferences and production should shift with the referential scene rather than reflecting only a global size-first prior.

### 2.3 Incremental Pragmatic Models

The Rational Speech Act (RSA) framework offers a natural way of formalizing these pressures. In RSA, a pragmatic speaker selects utterances by reasoning about a listener’s posterior, and the listener in turn reasons about the speaker’s choice (Goodman & Frank, 2016). Although standard RSA treats complete utterances as the units of choice, the same logic can be defined incrementally, with the speaker choosing one word at a time on the basis of how each token updates the listener’s beliefs (CohnGordon, Goodman, & Potts, 2019). Existing incremental pragmatic models and related work on redundancy establish that word-by-word reasoning can capture asymmetries between adjective positions and between prenominal and postnominal systems (Degen et al., 2020; Rubio-Fernández et al., 2021). What these accounts usually do not assume is that lexical meanings themselves change as the listener’s posterior changes. In most such accounts the posterior updates online, but the interpretation of an adjective remains fixed relative to the scene. Schlotterbeck and Wang (2023) introduced an alternative in which the threshold for a gradable adjective is recomputed from the listener’s current posterior at each stage of composition, together with the slider-rating paradigm that forms the basis for one of the experiments reported here. That study showed that the model qualitatively captures both a subjectivity-driven baseline ordering and a discriminatory-strength modulation, but it did not isolate the respective contributions of incremental production and context-updating semantics, nor did it assess their quantitative fit.

## 3 The Modelling Framework

We extend Schlotterbeck and Wang (2023) in four directions. We embed the model in a systematic  $2 \times 2$  comparison of speaker architecture and semantic regime, including a context-fixed ablation; we add a constrained production experiment that provides a richer

response space; we introduce a simulation study that identifies which ordering effects follow from context-updating composition alone; and we use Bayesian model comparison to quantify each mechanism’s contribution and to reveal an interaction that the earlier qualitative analysis could not detect. Here we describe the framework at a conceptual level; the meaning functions, speaker definitions, and inference setup are given formally in Section 5. The remaining subsections develop these components in turn.

### 3.1 Reference, Scenes, and the Speaker’s Problem

The model addresses a referential task. A scene contains several objects that vary in size, colour, and form, and one of them is the intended referent — in our materials, the largest object that matches the target colour and form. A speaker must choose an adjective string that lets a listener identify this referent, and the listener interprets that string by progressively narrowing the set of objects it could pick out. Because objects differ along three properties, the same referent can be described in many ways, and the order in which adjectives are produced is itself part of what the speaker chooses.

### 3.2 Two Mechanisms for Order Sensitivity

Determinate properties such as colour and form are treated as nearly categorical, whereas size is gradable: whether an object counts as “big” depends on the comparison class it is evaluated against, so the size term is informative only relative to the objects currently under consideration. Against this background, the framework isolates two ways in which adjective order can come to matter. The first is the *speaker architecture*. A global speaker evaluates complete adjective strings and chooses among them as wholes, whereas an incremental speaker chooses one adjective at a time, weighing at each step how the next word would reshape the listener’s beliefs. Incremental choice makes the usefulness of a modifier depend on where it occurs in the string, not only on the description as a whole. The second is the *semantic regime*. Under context-fixed semantics, the size threshold is computed once from the scene and held constant, so reordering the adjectives cannot change what “big” means. Under context-updating semantics, the threshold is recomputed from the listener’s

running beliefs, so that once “blue” has restricted attention to blue objects, “big” is judged against just those objects; “big blue” and “blue big” can then differ even though they contain the same words. This non-commutativity connects to Teodorescu’s (2006) observation that ordering restrictions weaken precisely when the alternative orders are not truth-conditionally equivalent.

Crossing these two binary choices yields a  $2 \times 2$  space of models — speaker architecture (global vs. incremental) by semantic regime (context-fixed vs. context-updating). We do not claim that production-level incrementality is required for adjective-ordering preferences in general; the subjectivity literature and our own simulations (Section 7) indicate that context-sensitive restrictive composition already accounts for much of the canonical tendency. The question is how much explanatory work remains once that baseline is in place, and in particular whether incremental production and context-updating semantics are independently useful or interact.

### 3.3 Qualitative Predictions

This framework yields qualitative predictions that the two experiments are designed to test (Table 1). A stable, subjectivity-aligned ordering prior yields a baseline size-first preference that survives even when colour alone identifies the referent (P1). The pragmatic component is sensitive to which property discriminates the referent in the current scene, so the size-first preference should weaken as colour becomes the more useful property (P2). Following the subjectivity hypothesis, higher conceptual blur of size — operationalised here as low size discriminability — should yield a stronger size-first preference, because a more subjective adjective benefits from sitting outside the less subjective one (P3) (Franke et al., 2019; Scontras et al., 2020). Finally, because production allows the full inventory of one-, two-, and three-adjective strings, the framework predicts a graded over-informativity gradient that the binary slider task cannot reveal, and predicts further that this gradient is itself sensitive to conceptual blur, with blurrier size displays eliciting longer, more redundant strings (P4).

**Table 1**

*Qualitative predictions of the modelling framework and where each is tested.*

|    | Prediction                                                                                                                                                                                                                      | Tested in  |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| P1 | A baseline size-first preference persists even when colour alone identifies the referent.                                                                                                                                       | Exp. 1 & 2 |
| P2 | The size-first preference weakens as colour becomes the more discriminating property.                                                                                                                                           | Exp. 1 & 2 |
| P3 | Following the subjectivity hypothesis, higher conceptual blur of size (low size discriminability) yields a stronger size-first preference.                                                                                      | Exp. 1 & 2 |
| P4 | Production exposes a graded over-informativity gradient across one-, two-, and three-adjective strings, and that gradient is itself modulated by conceptual blur: blurrier size displays elicit longer, more redundant strings. | Exp. 2     |

## 4 Experiments

We tested the qualitative predictions of the modelling framework (Table 1) with two complementary online referential tasks in German. Experiment 1 used graded slider judgments to measure how acceptable competing adjective orders are in a controlled scene. Experiment 2 used constrained production to measure which adjective sequences speakers actually choose in the same referential displays. Combining the two tasks allows us to separate stable ordering preferences from their online realization in production.

### 4.1 Shared Method

#### 4.1.1 Design

Both experiments used the same referential design. On every critical trial, participants saw six objects and had to refer to the framed target by its perceptual properties rather than by screen position. The target was always the largest object that

matched the intended colour and form. The analyses reported here focus on the theoretically central size–colour subset. Within that subset, referential context had three levels: size-sufficient (size alone identifies the target), colour-sufficient (colour alone identifies the target), and both-necessary (neither property alone suffices). A second factor, size discriminability, varied whether the size contrast between the target and its competitors was high or low. Referential context varied within participants; size discriminability was manipulated between participants (Table 2).

**Table 2**

*Shared design of the reported size–colour analyses*

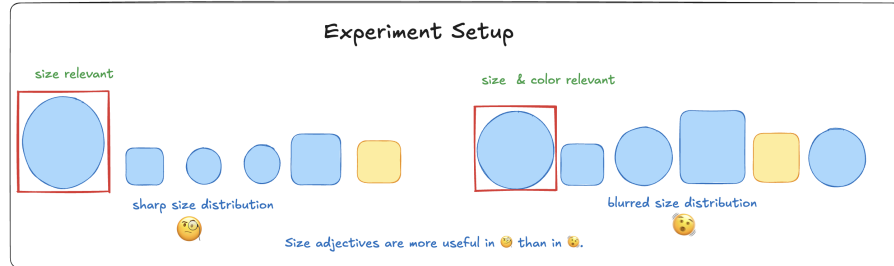
| Factor                                                                                                                     | Levels                                                |
|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Referential context                                                                                                        | Size-sufficient, colour-sufficient, both-necessary    |
| Size discriminability                                                                                                      | High (strong size contrast), low (weak size contrast) |
| <i>Fixed display property.</i> The referent is the largest, colour-matching, form-matching object in a six-object context. |                                                       |

The broader paradigm also included size–form and colour–form trials to widen the material base and reduce dependence on a single adjective pairing. These conditions remain part of the experimental design, but the present analysis concentrates on the size–colour subset because it provides the clearest contrast between a gradable adjective and a determinate feature adjective.

#### 4.1.2 *Materials*

The experiments were built from 27 underlying scene types. Each display contained six objects varying in size and in discrete colour and form values. The target was marked with a coloured frame (Figure 2). Critical vocabularies were fixed across the study, with six colour terms and six form terms reused throughout the scene set, and the modified noun was always *Aufkleber* (‘sticker’). Items were distributed across participants in a Latin-square

design so that each participant saw every scene type exactly once in each condition, and condition–item pairings were counterbalanced across participants. In the computational model, each display can be represented as a  $6 \times 3$  matrix  $(s, c, f)$  with the referent at index 0, but the experimental task presented these scenes as ordinary visual arrays rather than as explicit feature lists.



**Figure 2**

*Example experimental item. The referent (circled) is the largest, colour-matching, form-matching object. Panels illustrate high- and low-discriminability size conditions.*

#### 4.1.3 Procedure

Participants completed the experiments online via Prolific and were instructed to identify the framed target using its perceptual properties, not spatial descriptions such as *left* or *right*. The cover story framed each trial as a telephone conversation about collectible stickers: participants saw a set of six stickers on screen, one highlighted by a coloured frame, and were asked to refer to the target for a partner who could see the same stickers but neither the frame nor the spatial layout. This made spatial terms uninformative and forced reliance on adjective descriptions. Both experiments began with a consent form and a brief practice block (three training trials with feedback) that familiarised participants with the interface and the referential task. The main trials were divided into four blocks of approximately equal length, with self-paced breaks between blocks. Trial order within each block was randomised independently for each participant. The two experiments differed in response format: Experiment 1 presented two competing adjective orders on a continuous slider (180 trials total, drawn from one of six counterbalanced lists), whereas Experiment 2 asked participants

to type the adjective material they would place before the noun (135 trials, again from one of six lists, plus two additional dictionary-style practice items). Despite this difference, both tasks probed the same question: whether adjective ordering tracks baseline size-first preferences, referential context, and the discriminatory value of size in the current display.

## 4.2 Experiment 1: Slider-Rating Study

### 4.2.1 *Participants*

Participants ( $N = 134$  after exclusions; 70 low discriminability, 64 high discriminability) were recruited online via Prolific and reported German as their native or fluent language. Exclusion criteria were: more than three missing trial-list entries, completion times more than two standard deviations from the sample mean, more than three failures on control items, or more than three “none fits” responses on experimental items. The analysed dataset contained 10,606 critical observations. Mean completion time was 27.6 minutes. Participants were compensated via Prolific at a rate above the platform-recommended minimum.

### 4.2.2 *Procedure*

Experiment 1 implemented the shared referential design as a comparative judgment task. Participants were asked to imagine a referential exchange about sticker collections and to evaluate which description would be better for identifying the framed target. On each critical trial, two competing adjective orders appeared on opposite sides of a continuous slider, together with a “none of the descriptions fits” option. Left–right presentation of the two orders was randomised across trials. Each participant saw one of six lists built from the 27 underlying scenes, yielding 81 critical trials and 99 filler and control trials (180 total). The filler and control items discouraged fixed response strategies and ensured attention to the referential task. Raw slider values were recorded on a 0–100 scale and rescaled to  $[-0.5, 0.5]$ , with positive values indicating a size-first preference. The descriptive analysis used a linear mixed-effects model with centred slider rating as the dependent variable and referential context, size discriminability, and their interaction as fixed effects, with random intercepts

for participant and item. For the Bayesian model comparison, these ratings were modelled with a Zero-One-Inflated Beta (ZOIB) likelihood to accommodate boundary mass at 0 and 1.

### 4.2.3 Results

Figure 3 shows that size-first orderings were preferred across all three referential contexts, but the strength of the preference depended on which property was informative. Referential context produced a large main effect,  $F(2, 3349) = 213.7$ ,  $p < .001$ ;  $\chi^2(2) = 402.6$ ,  $p < .001$ . The size-first preference was strongest in size-sufficient displays, intermediate in both-necessary displays, and weakest in colour-sufficient displays. Notably, even colour-sufficient contexts did not produce a qualitative reversal: mean centred ratings remained above zero.

Size discriminability did not produce a reliable main effect,  $F(1, 132) = 0.03$ ,  $p = .86$ , but it interacted with referential context,  $F(2, 3349) = 5.84$ ,  $p = .003$ ;  $\chi^2(2) = 11.7$ ,  $p = .003$ . The interaction was asymmetric: higher discriminability slightly increased the size-first preference in size-sufficient contexts (low: 0.30, high: 0.34) but slightly reduced it in both-necessary (0.25 vs. 0.20) and colour-sufficient contexts (0.06 vs. 0.04). Post-hoc pairwise comparisons (Bonferroni-corrected) confirmed that no individual simple effect of discriminability within any referential context reached significance (all  $p > .12$ ), indicating that the interaction is driven by the directional pattern across contexts rather than by any single cell contrast (see Appendix for full pairwise results). This pattern is consistent with a discriminatory-strength interpretation: ordering preferences track the usefulness of size for identifying the target, but a baseline size-first tendency persists even when colour is locally more discriminating. P1 and P2 are confirmed; P3 is not. Size discriminability produces no main effect, no simple effect within any referential context reaches significance, and the asymmetric interaction reverses direction in size-sufficient displays, where higher discriminability slightly strengthens rather than weakens size-first ratings. The interaction is best read as a small context-bound modulation, not as a subjectivity-aligned main effect.



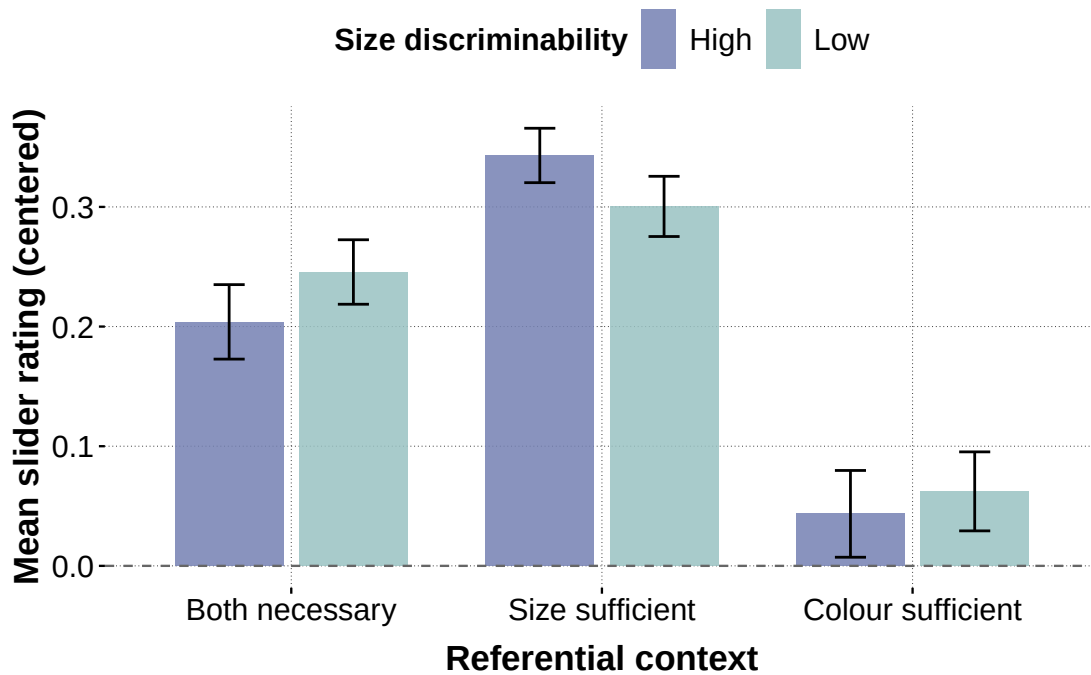


Figure 3

*Slider-rating data. Mean centred slider ratings by referential context and size discriminability, with 95% confidence intervals; positive values indicate size-first preference.*

### 4.3 Experiment 2: Free-Production Study

#### 4.3.1 Participants

Participants ( $N = 113$  after exclusions; 60 low discriminability, 53 high discriminability) were recruited online via Prolific; anyone who had completed the slider study was excluded. Further exclusion criteria were: completion times more than two standard deviations from the mean, more than 80% identical responses, or more than 10% missing or unannotatable responses. The analysed dataset contained 9,639 critical observations. Mean completion time was 32.4 minutes. Compensation followed the same Prolific rate as Experiment 1.

#### 4.3.2 Procedure

Experiment 2 used the shared referential design in a constrained production format. Participants saw the framed target together with a sentence prompt and typed the adjective

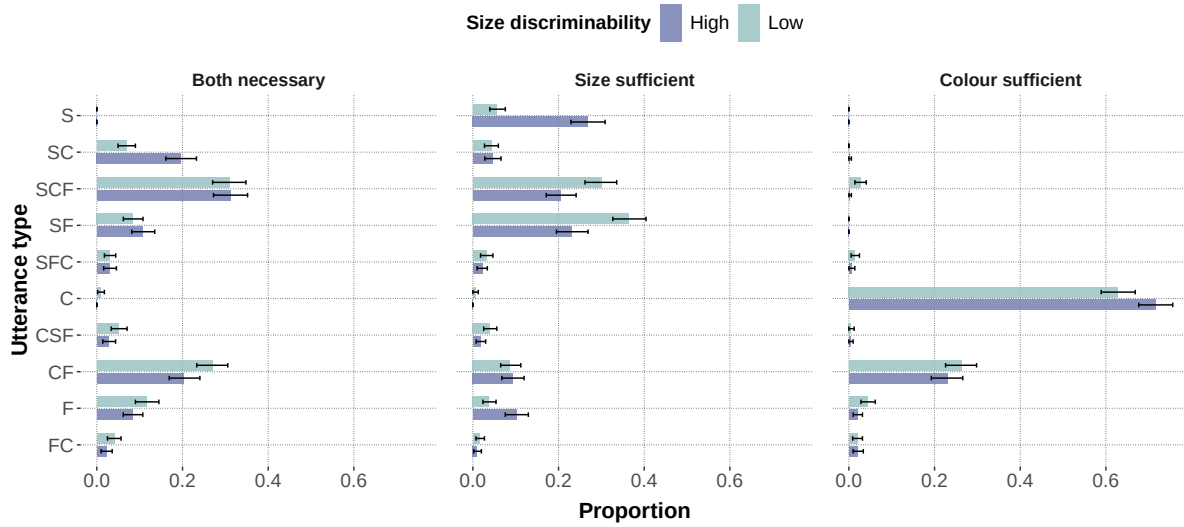
material needed before the noun *Aufkleber*. Restricting the response to the adjective slot rather than eliciting a full sentence reduced off-task variability and provided a controlled basis for coding. Instructions again prohibited spatial descriptions, and a practice block familiarised participants with the relevant colour and form vocabulary before the critical trials began. Each list contained 81 critical trials; the filler load was reduced relative to Experiment 1 (135 total trials vs. 180) to accommodate the greater time burden of typed production.

Responses were coded into 15 utterance categories corresponding to all one-, two-, and three-adjective strings built from size, colour, and form terms (3 one-adjective, 6 two-adjective, 6 three-adjective). Misspellings and transparent lexical alternatives were normalised when their semantic class was unambiguous; spatial descriptions, comparative or ordinal size expressions, and other unclassifiable responses were excluded. The dependent variable for the Bayesian model comparison is the categorical utterance label  $y_i \in \{1, \dots, 15\}$  for each trial  $i$ . For descriptive statistics and targeted mixed-effects analyses, we also tracked two derived binary outcomes: whether an utterance was size-initial and whether it was over-informative (i.e., a three-adjective response in both-necessary contexts, or a multi-adjective response in size-sufficient or colour-sufficient contexts).

### 4.3.3 Results

The production data show stronger context sensitivity than the slider ratings. As Figure 4 illustrates, size-first utterances dominated in size-sufficient contexts, colour-first utterances dominated in colour-sufficient contexts, and both-necessary contexts fell between the two extremes. A mixed-effects logistic regression on size-initial responses yielded a strong main effect of referential context,  $\chi^2(2) = 1859.7$ ,  $p < .001$ , no reliable main effect of size discriminability,  $\chi^2(1) = 0.79$ ,  $p = .37$ , and a reliable interaction,  $\chi^2(2) = 31.4$ ,  $p < .001$ . As in the slider data, the interaction was asymmetric: higher discriminability reduced the size-first advantage in size-sufficient contexts but increased it in both-necessary contexts.

A complementary analysis of over-informativity reveals a related efficiency pressure.

**Figure 4**

*Free-production data. Utterance-type proportions by referential context and size discriminability, with bootstrapped 95% confidence intervals; only utterance types reaching at least 2% in one condition are shown.*

Over-informative utterances were most frequent in size-sufficient contexts and least frequent in colour-sufficient contexts, producing a strong main effect of referential context,  $\chi^2(2) = 725.7, p < .001$ . Unlike size-initiality, over-informativity also showed a main effect of size discriminability,  $\chi^2(1) = 10.3, p = .001$ , with more multi-adjective responses in low- than in high-discriminability displays. This effect interacted with referential context,  $\chi^2(2) = 63.8, p < .001$ , driven primarily by size-sufficient contexts. The pattern aligns with the broader communicative-efficiency literature on redundancy: speakers produce additional material when it helps resolve reference, even if it is not strictly necessary for uniqueness. These results carry P1 and P2 into unconstrained production: the size-first baseline (P1) is attenuated where colour alone identifies the referent, and the context modulation (P2) emerges more sharply than in the slider task. P3 is again not supported. The interaction on size-initial responses is asymmetric, with higher discriminability reducing size-first only in size-sufficient contexts and increasing it in both-necessary contexts, so blur does not monotonically promote size-first ordering. P4 is confirmed: the richer response space reveals

a graded over-informativity gradient across one-, two-, and three-adjective strings, and blurrier displays elicit more redundant strings overall.

#### 4.4 Interim Summary

Across both experiments, three empirical regularities emerge. First, there is a robust baseline preference for size-first orderings that persists even when colour is the only discriminating property. Second, this preference is modulated by referential context: size-first choices increase when size is informative and decrease when colour is more useful, consistent with a discriminatory-efficiency account. Third, size discriminability does not produce reliable main effects, and the direction of its interaction with referential context varies across cells: in the slider task, higher discriminability raised size-first ratings in size-sufficient displays but lowered them in both-necessary and colour-sufficient displays, while in production the directions were reversed for size-sufficient and both-necessary displays. The data therefore run against the subjectivity-aligned prediction that blurrier size yields a uniformly stronger size-first preference. The production data amplify the first two patterns relative to the slider data: free production elicits stronger context sensitivity and additionally reveals an over-informativity gradient that is itself sensitive to size blur, with blurrier displays eliciting longer, more redundant strings. This context-sensitivity mirrors the visual-world finding that listeners interpret prenominal modifiers incrementally and use them to restrict the referential domain to contrastive objects well before the noun is encountered (Sedivy, Tanenhaus, Chambers, & Carlson, 1999; Tanenhaus, Spivey-Knowlton, Eberhard, & Sedivy, 1995). The present production data complement that picture from the speaker side: ordering and over-informativity track which property is discriminating in the current scene, suggesting that speakers shape utterances around the same incremental restriction that listeners exploit. Together, these results motivate a formal model that can capture both the stable ordering bias and its context-dependent modulation, while accommodating the richer response space of unconstrained production.

## 5 Computational Model

This section formalizes the modelling framework introduced in Section 3. It specifies the literal-listener semantics, the global and incremental speakers, the context-fixed and context-updating regimes, and the inference setup for the  $2 \times 2$  model comparison. The experiments were broader than the fitted model comparison: the full paradigm included size–form and colour–form trials, but the analyses reported here focus on size–colour scenes, where the theoretical contrast between a gradable adjective and a determinate feature adjective is clearest. Form therefore remains part of the general representational vocabulary and of the production response space, but it is not the primary mechanistic contrast in the current fit.

### 5.1 Representational Primitives

An object is a triple  $o = (s, c, f)$ , where  $s \in [1, 10]$  denotes size,  $c \in \{0, 1\}$  indicates colour match, and  $f \in \{0, 1\}$  indicates form match. A context  $\mathcal{C}$  is a set of  $n_{\text{obj}}$  objects. In the experiments, the intended referent is always the largest object with  $c = f = 1$ .

Utterances are built from three adjective types, S, C, and F. An utterance  $u = (a_1, \dots, a_T)$  is an ordered sequence of adjectives  $a_t \in \{\text{S}, \text{C}, \text{F}\}$  with  $T \in \{1, 2, 3\}$ . Combining all lengths and orderings yields 15 surface forms:  $\{\text{S}, \text{SC}, \text{SCF}, \text{SF}, \text{SFC}, \text{C}, \text{CS}, \text{CSF}, \text{CF}, \text{CFS}, \text{F}, \text{FS}, \text{FSC}, \text{FC}, \text{FCS}\}$ . This full inventory reflects the broader experimental vocabulary. For the reported model fits, however, the scene subset is size–colour: C denotes the determinate feature that competes with size for identification, while F remains available chiefly as an additional, potentially redundant modifier in production.

### 5.2 Meaning Functions

**Context-dependent size semantics.** We model size adjectives with a context-sensitive threshold semantics following Schmidt, Goodman, Barner, and Tenenbaum (2009), who showed that the interpretation of gradable adjectives depends on the statistics of the comparison class. Franke et al. (2019) adopted the same approach in their simulation of

adjective-ordering preferences. A threshold is computed from the current comparison class  $\mathcal{C}$ :

$$\theta_k(\mathcal{C}) = x_{\max}(\mathcal{C}) - k(x_{\max}(\mathcal{C}) - x_{\min}(\mathcal{C})), \quad (1)$$

where  $x_{\min}(\mathcal{C})$  and  $x_{\max}(\mathcal{C})$  are the lower and upper bounds of the mid-range of the contextual size distribution and  $k \in [0, 1]$  controls how permissively “big” is applied. The resulting semantic value  $\llbracket S \rrbracket(o, \mathcal{C}) \in [0, 1]$  is graded rather than binary: an object’s size is compared to the threshold via a normal slack function scaled by a perceptual-blur parameter  $w_f$ , consistent with scalar variability in the Approximate Number System (Feigenson, Dehaene, & Spelke, 2004). This context dependence is what makes the context-fixed vs. context-updating distinction possible: because  $\theta_k$  depends on  $\mathcal{C}$ , changing which objects are in the comparison class changes what counts as big.

**Colour and form semantics.** Colour and form are in principle determinate: an object either matches the feature or it does not. Following the relaxed-semantics approach of Degen et al. (2020) and Waldon and Degen (2021), however, we soften these denotations by parameterizing them with  $\nu \in (0, 1)$ , which can be interpreted as the probability that a feature-matching object is correctly described by the adjective. This relaxation allows even determinate properties to carry graded information, which is important for modeling redundant modification. In the reported model fits, colour is the primary determinate comparison case because inference is carried out on size–colour scenes, whereas form is retained as a general determinate feature term and as a possible redundant modifier in production. In the slider models, colour and form share a single  $\nu$  parameter. In the production models, they receive separate fixed values,  $\nu_{\text{colour}}$  and  $\nu_{\text{form}}$ , obtained from a preliminary inference run and held constant across all model variants.

### 5.3 Incremental Listener

The literal listener constructs utterance meanings incrementally by combining token-level semantic contributions while updating the effective comparison class. Let  $u = (w_1, \dots, w_T)$  be an utterance. The listener maintains a sequence of belief states

$\{P_t(o)\}_{t=0}^T$ , where  $P_t(o)$  denotes the posterior belief after incorporating the suffix  $(w_{t+1}, \dots, w_T)$ :

$$P_t(o) = P(o \mid w_{t+1}, \dots, w_T), \quad t = 0, \dots, T, \quad (2)$$

with base case

$$P_T(o) = P_{\text{prior}}(o). \quad (3)$$

For  $t = T, T-1, \dots, 1$ , beliefs are updated according to

$$P_{t-1}(o) \propto P_t(o) \llbracket w_t \rrbracket(o), \quad (4)$$

which normalizes to

$$P_{t-1}(o) = \frac{P_t(o) \llbracket w_t \rrbracket(o)}{\sum_{o'} P_t(o') \llbracket w_t \rrbracket(o')}. \quad (5)$$

After all tokens have been processed, the listener arrives at

$$P_0(o) = P(o \mid u). \quad (6)$$

The composition rule in Equation (4) is shared by both semantic regimes of the model. What differs is how the meaning function  $\llbracket w_t \rrbracket$  is computed for gradable adjectives such as size.

**Context-updating semantics.** Under the context-updating regime, the size threshold  $\theta_k$  is recomputed from the listener's current belief state at each composition step:

$$\llbracket w_t \rrbracket_{\text{UPD}}(o) = \llbracket w_t \rrbracket_{P_t}(o), \quad (7)$$

where  $P_t$  is the posterior after processing all words to the right of position  $t$ . The current belief state therefore plays a dual role: it serves as the prior for the next update step, and it determines the semantics of the next token. Because  $P_t$  itself depends on the semantics of later words, this creates a feedback loop:

$$\llbracket w_t \rrbracket_{P_t} \text{ depends on } P_t(o) \propto P_{t+1}(o) \llbracket w_{t+1} \rrbracket_{P_{t+1}}(o). \quad (8)$$

Word order changes the sequence of comparison classes and thereby the interpretation of the utterance, yielding non-commutativity:

$$P(o \mid w_1, w_2) \neq P(o \mid w_2, w_1) \quad \text{in general.} \quad (9)$$

**Context-fixed semantics.** Under the context-fixed regime, the size threshold is computed once from the scene prior and held constant throughout composition:

$$\llbracket w_t \rrbracket_{\text{FIX}}(o) = \llbracket w_t \rrbracket_{P_{\text{prior}}}(o) \quad \text{for all } t. \quad (10)$$

The listener’s beliefs still update incrementally via Equation (4), but this update no longer feeds back into the meaning function. For deterministic adjectives such as colour and form, both regimes are identical because their semantics does not depend on the comparison class. The context-fixed regime therefore isolates the contribution of incremental belief updating from the additional effect of posterior-dependent meaning.

#### 5.4 Utterance Prior and Speaker Models

The model incorporates a prior over utterance forms that is separate from the pragmatic component. This prior captures stable ordering tendencies — such as the well-documented subjectivity-driven size-first preference — that the speaker brings to the task before any scene-specific pragmatic reasoning.

In the slider-rating models, the prior is minimal: a single inferred bias parameter  $b \geq 0$  that penalizes the reverse (colour-first) order relative to the canonical (size-first) order. Because  $b$  is estimated from data rather than fixed, the posterior indicates how much residual ordering preference remains after the RSA mechanism has been applied; if the pragmatic model were sufficient on its own, the posterior for  $b$  would concentrate near zero.

In the production models, the prior is richer: a language-model distribution  $P_{\text{LM}}(u)$  over the 15 utterance forms, derived from German GPT-2 surprisal estimates averaged across the experimental adjective–noun combinations. The contribution of this prior is scaled by an inferred parameter  $\beta$ , yielding utterance cost  $\text{cost}_{\text{LM}}(u) = -\beta \log P_{\text{LM}}(u)$ . Like  $b$  in the slider model, the LM prior encodes stable word-order tendencies from language experience, which the pragmatic speaker then modulates in light of the current scene.



**Global speaker.** The global speaker computes a distribution over complete utterances by applying a softmax to utterance-level utility:

$$S_{\text{gb}}(u \mid o) = \frac{\exp(U(o, u))}{\sum_{u'} \exp(U(o, u'))}. \quad (11)$$

For the production model, utility is defined as

$$U(o^*, u) = \alpha \log L(o^* \mid u) + \beta \log P_{\text{LM}}(u), \quad (12)$$

where  $\alpha$  is the RSA rationality parameter. In the slider-rating models, the utility simplifies to

$$U(o, u) = \alpha (\log L(o \mid u) - b_u), \quad (13)$$

where  $b_u = 0$  for the size-first order and  $b_u = b$  for the reverse order.

**Incremental speaker.** The incremental speaker chooses one token at a time. For an intended referent  $o^*$ ,

$$S_{\text{inc}}(u \mid o^*) = \prod_{t=1}^T S_t(w_t \mid u_{<t}, o^*), \quad u_{<t} = (w_1, \dots, w_{t-1}), \quad (14)$$

with local decision rule

$$S_t(a \mid u_{<t}, o^*) \propto [L(o^* \mid u_{<t} + a)]^\alpha. \quad (15)$$

Combining local pragmatic choice with utterance-level cost yields, for the production model,

$$S_{\text{inc}}(u \mid o^*) \propto P_{\text{LM}}(u)^\beta \cdot \prod_{t=1}^T \frac{[L(o^* \mid u_{<t} + w_t)]^\alpha}{Z_t(u_{<t}, o^*)}, \quad (16)$$

where

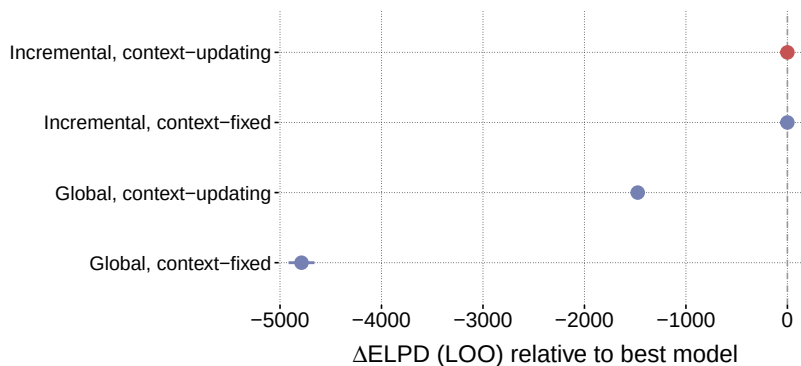
$$Z_t(u_{<t}, o^*) = \sum_a [L(o^* \mid u_{<t} + a)]^\alpha \quad (17)$$

ensures local normalization. In the slider-rating models, the LM prior is replaced by the bias term:  $S_{\text{inc}}(u \mid o^*) \propto \exp(-b_u) \cdot \prod_t S_t(w_t \mid u_{<t}, o^*)$ . The theoretical contrast is that the global speaker applies pragmatic reasoning once over complete utterances, whereas the incremental speaker applies it at each token position. When semantics is context-dependent, these two model variants are not equivalent.

## 6 Bayesian Model Comparison

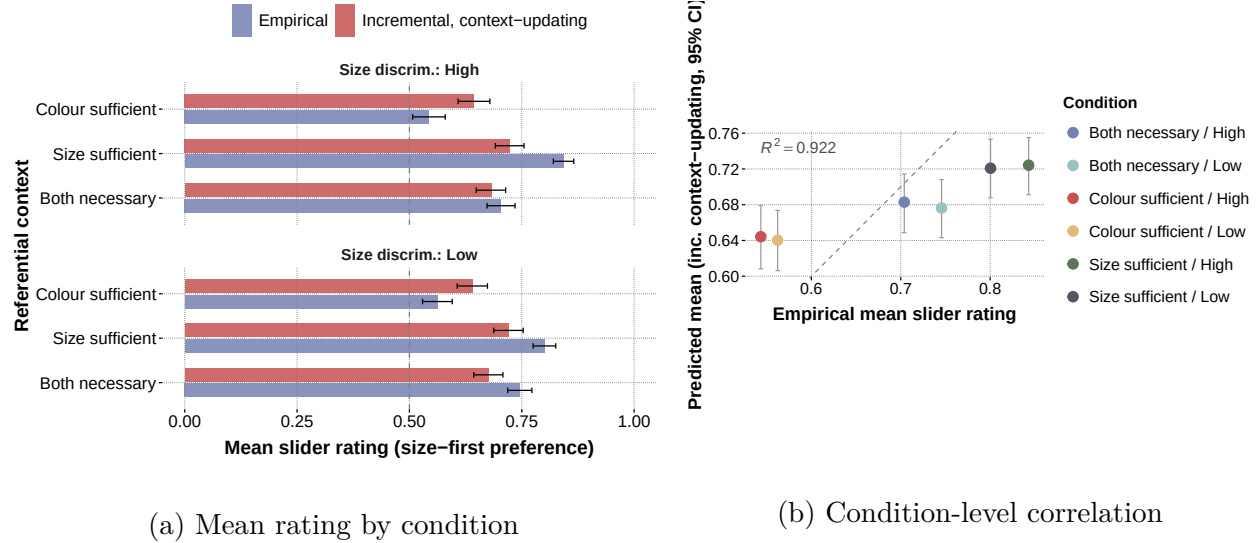
The  $2 \times 2$  comparison crosses the two mechanisms introduced in Section 3.2 — speaker architecture (global vs. incremental) and semantic regime (context-fixed vs. context-updating) — and is fitted independently to both datasets with hierarchical participant pooling. Because the four models in each dataset differ only structurally, fit differences isolate the architecture and the regime rather than expressive flexibility. We use Pareto-smoothed importance-sampling leave-one-out cross-validation (PSIS-LOO; Vehtari et al., 2017), reporting differences in expected log pointwise predictive density ( $\Delta\text{ELPD}$ ) relative to the best-fitting model together with the difference standard error (dse); a ratio  $|\Delta\text{ELPD}|/\text{dse}$  substantially greater than 1 indicates a significant difference in out-of-sample predictive accuracy. All models were implemented in NumPyro (Phan, Pradhan, & Jankowiak, 2019) and estimated with four MCMC chains using the NUTS sampler (Hoffman & Gelman, 2014); convergence was confirmed via  $\text{split-}\hat{R} < 1.01$  and effective sample sizes exceeding 400 for all parameters across all models. ?? summarises the full PSIS-LOO comparison; dataset-specific parameter inventories, likelihoods, and sampler settings are introduced with each subsection below.

### 6.1 Slider-Rating Study



**Figure 5**

*Slider-rating model comparison:  $2 \times 2$  speaker  $\times$  semantics.  $\Delta\text{ELPD}$  relative to the best model, with  $\pm 1$  difference standard error.*



**Figure 6** (a) Mean rating by condition (b) Condition-level correlation

*Posterior-predictive fit of the incremental, context-updating slider model. (a) Observed (blue) vs. model (red) mean slider rating by referential context  $\times$  size discriminability; dashed line marks 0.5 (no preference), error bars are 95% confidence and posterior-predictive intervals. (b) Observed vs. predicted condition means; dashed line is identity.*

The slider  $2 \times 2$  is fitted on a minimal speaker with a single rationality parameter  $\alpha$ , a bias intercept  $b$ , and a participant-level offset, with the semantic constants held fixed at  $k = 0.5$ ,  $w_f = 1.0$ , and  $\nu = 0.80$ . Slider responses cluster at the boundaries (0 and 1) and are modelled with a Zero-One-Inflated Beta likelihood; each chain drew 500 warmup and 500 post-warmup samples (target acceptance probability = 0.90, maximum tree depth = 8).

Figure 5 shows the  $2 \times 2$  comparison for the slider data. The dominant contrast is speaker architecture: both incremental models fit the data well and are statistically indistinguishable ( $\Delta\text{ELPD} = -0.06$ ,  $\text{dse} = 0.18$ ,  $|\Delta\text{ELPD}|/\text{dse} = 0.32$ ), whereas both global models perform far worse ( $\Delta\text{ELPD} > 1,400$ ). Within the incremental pair, context-updating semantics confers no detectable advantage over context-fixed semantics. For graded preference judgments between two competing orders, word-by-word pragmatic reasoning is therefore the dominant mechanism: it provides strong order sensitivity regardless of whether the size threshold is recomputed from the running posterior. Posterior-predictive checks

confirm that the incremental context-updating model captures the condition-level pattern well, tracking the empirical ordering across all six referential context  $\times$  size-discriminability cells with  $r = 0.96$  (squared Pearson  $\approx 0.92$ ; Figure 6). The model slightly under-predicts the strongest size-first conditions (size-sufficient) and slightly over-predicts the weakest (colour-sufficient), compressing the predicted range relative to the empirical range while preserving the rank-ordering. For the best-fitting incremental context-updating model, the posterior rationality parameter was  $\alpha = 0.69$  ( $SD = 0.06$ , 94% HDI  $[0.57, 0.79]$ ); the implied Beta concentration of the continuous ZOIB component was  $\varphi = 1/\sigma^2 \approx 1.93$  (94% HDI  $[1.85, 2.04]$ ), corresponding to  $\text{Beta}(\mu \cdot 1.93, (1 - \mu) \cdot 1.93)$  — a broad, near-uniform spread around the conditional mean that places most of the conditional structure in  $\mu$  and in the boundary inflations; and the hierarchical spread of participant-level rationality offsets was  $\tau = 0.12$  ( $SD = 0.01$ , 94% HDI  $[0.10, 0.14]$ ). Crucially, the bias intercept was  $b = 0.58$  ( $SD = 0.06$ , 94% HDI  $[0.47, 0.69]$ ), with posterior probability 0.91 above 0.50. This positive bias means that even after the RSA mechanism accounts for context-dependent informativeness, the model infers a residual size-first preference — consistent with the view that a stable ordering default persists beyond what discriminatory utility alone explains. The binary slider response space does not strongly discriminate between sharper and softer colour predicates: the slider posterior for  $\nu$  is broad (0.63, 94% HDI  $[0.50, 0.82]$ ), bumps the prior boundary at 0.5, only just excludes 0.80, and the four models remain statistically indistinguishable in PSIS-LOO across this range (all  $|\Delta\text{ELPD}|/\text{dse} < 1$ ; full numbers in Appendix E). The positive-bias finding above is therefore conditional on the fixed  $\nu = 0.80$  specification: under softer colour predicates the rationality  $\alpha$  and the bias  $b$  shift substantially (towards 1.5 and below 0.5, respectively). Within the global pair, context-updating semantics substantially outperforms context-fixed semantics ( $\Delta\text{ELPD} \approx 3,313$ ,  $|\Delta\text{ELPD}|/\text{dse} > 24$ ). Context-updating composition thus provides a minimal source of order sensitivity that partially compensates for the lack of incrementality, but both global models remain far worse than either incremental variant.

## 6.2 Free-Production Study

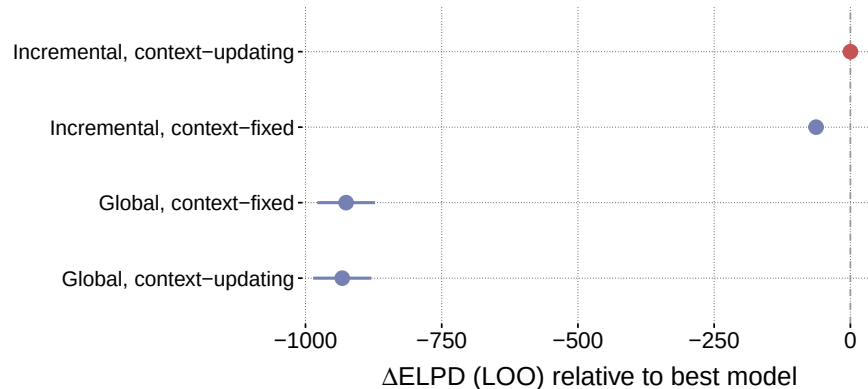
As the slider results above demonstrate, a minimal speaker — a single rationality parameter  $\alpha$ , the language-model weight  $\beta$ , and a participant-level offset — already reproduces the slider condition-level pattern almost perfectly. Free production is a fifteen-way choice over one-, two-, and three-adjective strings, and warrants a richer speaker. We extend the incremental speaker with a small set of theoretically motivated components, selected through an explicit parsimony criterion rather than tuned to residuals.

**Theoretically motivated components.** Each addition targets a documented property of production or of the adjective-ordering literature, not a particular residual cell. In the incremental speaker, size and colour receive separate informativity weights ( $\alpha_S$ ,  $\alpha_C$ ) because the two adjective types differ in how reliably they discriminate; the global speaker reasons over complete utterances and therefore uses a single rationality. A canonical-order penalty ( $\lambda_{\text{noncanon}}$ ) encodes the cross-linguistically robust colour-before-form tendency (Cinque, 1994; Scott, 2002; Sproat & Shih, 1991b). A sufficiency boost ( $\lambda_{\text{suff}}$ ) and a context-gated form-mention boost ( $\lambda_{\text{form-mod}}$ ) capture, respectively, the tendency to lead with the property that already individuates the referent and the additional form mention when neither size nor colour suffices. A graded length bias ( $\gamma_{\text{base}}$ ,  $\gamma_{\text{one}}$ ,  $\gamma_{\text{sharp}}$ ) controls the production of longer or shorter strings, including extra modification under blurry size. A lapse rate ( $\epsilon$ ) absorbs inattentive responses rather than distorting the pragmatic parameters. Finally, the determinate feature semantics are softened and anchored: colour and form are given graded denotations (as in the slider model) with empirically calibrated reliabilities.

The parsimony walk floors at ten parameters, with every retained coefficient’s 94% HDI excluding zero (Appendix C). The production data identify  $\nu_{\text{colour}}$  tightly at 0.59 [0.56, 0.62]; the other three semantic constants are not identifiably moved by the data.

The soft colour predicate is itself a substantive result. At  $\nu_{\text{colour}} \approx 0.59$  the colour informativity weight is roughly threefold larger and the lapse rate roughly halved relative to a near-deterministic specification. A near-deterministic colour denotation would suppress the

pragmatic weight on colour and force the resulting misfit into uniform noise; the calibrated graded predicate lets the informativity weights be estimated properly and lets far less of the data be explained away as inattention. The data therefore prefer a colour adjective that is graded rather than categorical, consistent with the relaxed-semantics motivation for  $\nu$ .

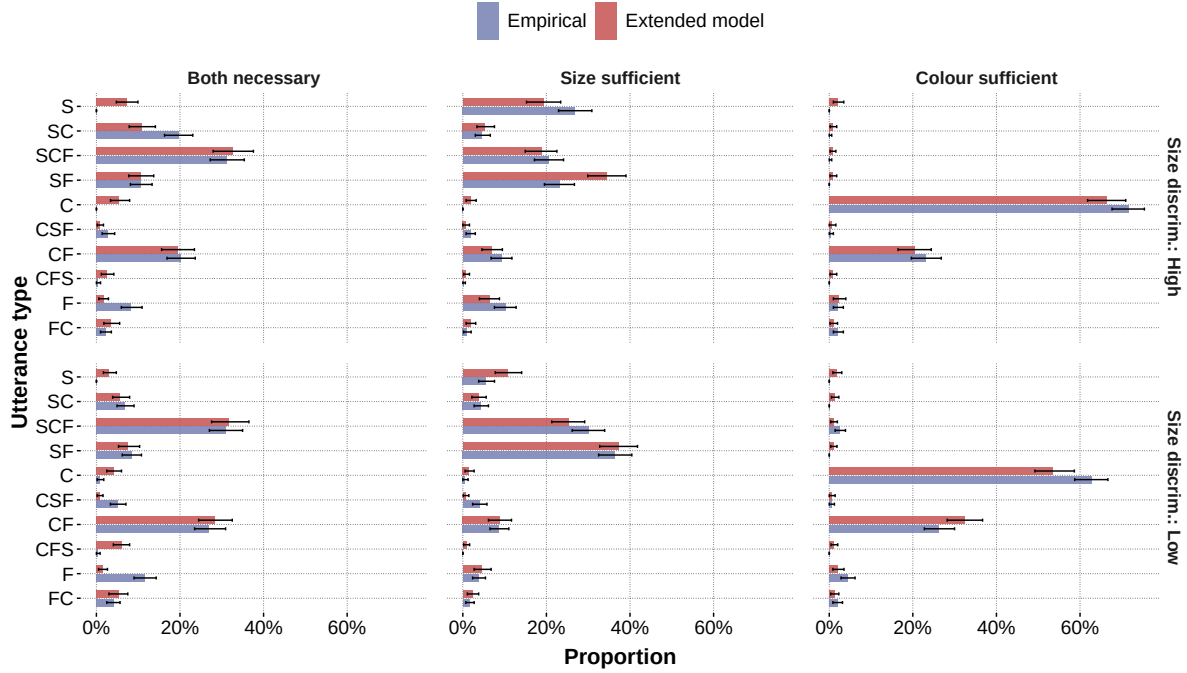


**Figure 7**

*Free-production model comparison: PSIS-LOO  $\Delta$ ELPD relative to the best model for the parameter-matched  $2 \times 2$  (speaker  $\times$  semantics), with  $\pm 1$  difference standard error.*

The resulting production speaker has ten population parameters, the participant hierarchy, and five empirically calibrated constants ( $\nu_{\text{colour}} = 0.59$ ,  $\nu_{\text{form}} = 0.50$ ,  $k = 0.50$ ,  $w_f = 0.69$ , and the fixed LM temperature); its full parameter inventory and priors are given in Table B1. The four models of the  $2 \times 2$  are parameter-matched on this inventory — identical ten-parameter set, identical participant hierarchy, identical calibrated constants — so any fit difference between models reflects the architecture and the semantic regime rather than expressive flexibility. Production responses are categorical over the fifteen utterance types; each chain drew 5,000 warmup and 2,000 post-warmup samples (target acceptance probability = 0.85, maximum tree depth = 7). The advocated production model is the incremental, context-updating variant; it reproduces the production data closely ( $R^2 \approx 0.94$ ,  $r = 0.97$ ; Figure 8 shows the by-condition fit and Figure 9 the condition-level correlation).

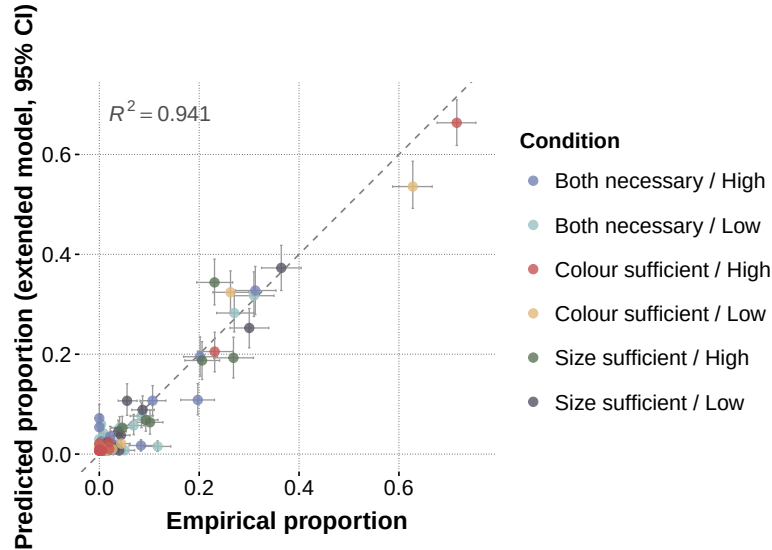
The dominant effect is speaker architecture. Both incremental models outperform both global models by a large margin: the speaker main effect on out-of-sample fit is +897

**Figure 8**

*Posterior-predictive utterance-type proportions for the incremental, context-updating production model. Observed (blue) vs. model (red) proportions by referential context  $\times$  size discriminability; error bars are 95% bootstrap and posterior-predictive intervals.*

ELPD, and the global models drop from  $R^2 \approx 0.94$  to  $R^2 \approx 0.78$ . This effect is not a flexibility artifact. Under the identical nominal inventory, the effective number of parameters ( $p_{\text{LOO}}$ ) diverges sharply: the incremental models keep the participant hierarchy identifiable ( $p_{\text{LOO}} \approx 145\text{--}175$  of the 113 participant offsets are data-informed), whereas the global models collapse it ( $p_{\text{LOO}} \approx 7$ ; the global speaker is too rigid for the data to inform per-participant rationality). The winning architecture therefore carries the *higher* realized complexity penalty and still wins by  $\sim 900$  ELPD — its advantage is despite, not because of, flexibility.

Within the incremental pair, the semantic regime now matters: the context-updating model significantly outperforms the context-fixed model ( $\Delta\text{ELPD} = 63$ ,  $\text{dse} = 14$ ,  $|\Delta\text{ELPD}|/\text{dse} = 4.5$ ). Within the global pair, the two regimes are effectively tied, with context-updating if anything slightly worse ( $\Delta\text{ELPD} = 7$ ,  $\text{dse} = 2.6$ ). The speaker  $\times$

**Figure 9**

*Observed vs. predicted utterance-type proportions across all condition  $\times$  utterance cells for the incremental, context-updating production model; dashed line is identity.*

semantics interaction (+70 ELPD) is the theoretically informative quantity:

context-updating threshold computation adds predictive value, but only when paired with an incremental architecture that can exploit non-commutative composition at each step. The parameter-matched  $2 \times 2$  comparison is shown in Figure 7, and the model’s posterior-predictive fit in Figures 8 and 9. Production is thus precisely the setting where word-by-word planning and context-updating semantics jointly matter, consistent with the view that adjective ordering in production reflects online discriminatory utility rather than utterance-level well-formedness alone.

The posterior parameters of the advocated (incremental, context-updating) model are reported in full in Table B1; every 94% HDI excludes zero. The size and colour informativity weights are well separated ( $\alpha_S \approx 8.0$ ,  $\alpha_C \approx 3.8$ ), the canonical colour-before-form penalty is large and tightly estimated ( $\lambda_{\text{noncanon}} \approx 4.4$ ), and the lapse rate is modest ( $\epsilon \approx 0.11$ ), so little of the fit is absorbed by uniform noise. The participant-level spread is substantial ( $\tau \approx 1.4$ ), consistent with the high effective parameter count and with genuine individual variability in



pragmatic engagement under incremental production.

Three additional ablations of the core incremental speaker are reported in the Appendix. A lookahead variant, which evaluates both remaining adjective orders before committing to a word, performed worse than the standard greedy incremental model, indicating that local step-by-step planning suffices for this task. An LM-only ablation, which removes the RSA component and relies solely on the corpus-derived prior, likewise fell short, confirming that the utterance prior alone cannot account for context-sensitive ordering. An RSA-only ablation, which removes the LM prior, failed the PSIS-LOO diagnostic entirely (high Pareto- $k$  values), suggesting that purely pragmatic choice without a lexical prior produces degenerate predictive distributions for this response space.

### 6.3 Cross-Dataset Summary

Comparing the two datasets reveals a diagnostic interaction between speaker architecture and semantic regime. Within the incremental models, context-updating semantics significantly improves fit for production ( $|\Delta\text{ELPD}|/\text{dse} = 4.5$ ) but not for slider ratings ( $|\Delta\text{ELPD}|/\text{dse} = 0.32$ ). Within the global models, the complementary pattern holds: context-updating semantics provides a massive advantage in the slider data ( $\Delta\text{ELPD} \approx 3,313$ ) but confers no benefit in production (a slight decrement,  $|\Delta\text{ELPD}|/\text{dse} = 2.6$ ). This cross-pattern suggests that context-updating composition is most useful when the model otherwise lacks a source of order-relevant information: in the global slider case, recursive threshold updating is the only mechanism that can break the symmetry between orders; in production, the LM prior already provides ordering information, making the global recursive advantage redundant. Conversely, when the speaker architecture itself introduces order sensitivity through incremental decision-making, context-updating semantics adds value only when the response space is rich enough (15-way production) to reward finer-grained discriminatory computation. The same response-space asymmetry is reflected in how strongly each dataset identifies the colour reliability  $\nu$ : production tightly identifies  $\nu_{\text{colour}}$  at 0.59 (94% HDI [0.56, 0.62]), with substantive consequences for the other

parameters, whereas the slider posterior for  $\nu$  is broad and bumps the prior boundary (0.63, HDI [0.50, 0.82]) with negligible ELPD consequences across the  $2 \times 2$ . The binary preference response space is therefore informative about speaker architecture but not about the sharpness of the colour predicate, while the 15-way production response space disambiguates both.

All models reported in the main text use hierarchical participant-level pooling (random intercepts on the rationality parameter). Pooled variants, which estimate a single set of parameters across all participants, were consistently worse and are reported in the Appendix.

## 7 Simulation Study

### 7.1 Goals and Design

The simulation addresses three questions that the model comparison alone cannot answer. First, does the model *generate* a size-first ordering preference across a broad range of contexts, or does this pattern emerge only after fitting to data? Second, which parameter regions produce a size-first advantage and which, if any, produce a colour-first reversal? Third, how do speaker architecture and semantic regime jointly determine the ordering preference — in particular, is context-updating semantics necessary for the baseline effect, or does it emerge from incremental production alone?

To answer these questions, we ran a full factorial sweep over the parameters listed in Table 3, crossing both speaker models (incremental, global) with both semantic regimes: recursive (the context-updating regime of Equation (7)) and static (the context-fixed regime of Equation (10)). Under recursive semantics, the size threshold adapts to the literal listener’s running posterior at each composition step; under static semantics, the threshold is computed once from the uniform prior and held fixed. For each parameter combination, 1,000 random six-object contexts were sampled, yielding 12,000,000 simulated trials in total. On each trial, the simulation computed the speaker’s probability of producing the size-first vs. colour-first ordering for the target referent and recorded the size-first advantage (the

difference in communicative success between the two orders).

**Table 3**

*Simulation parameter grid*

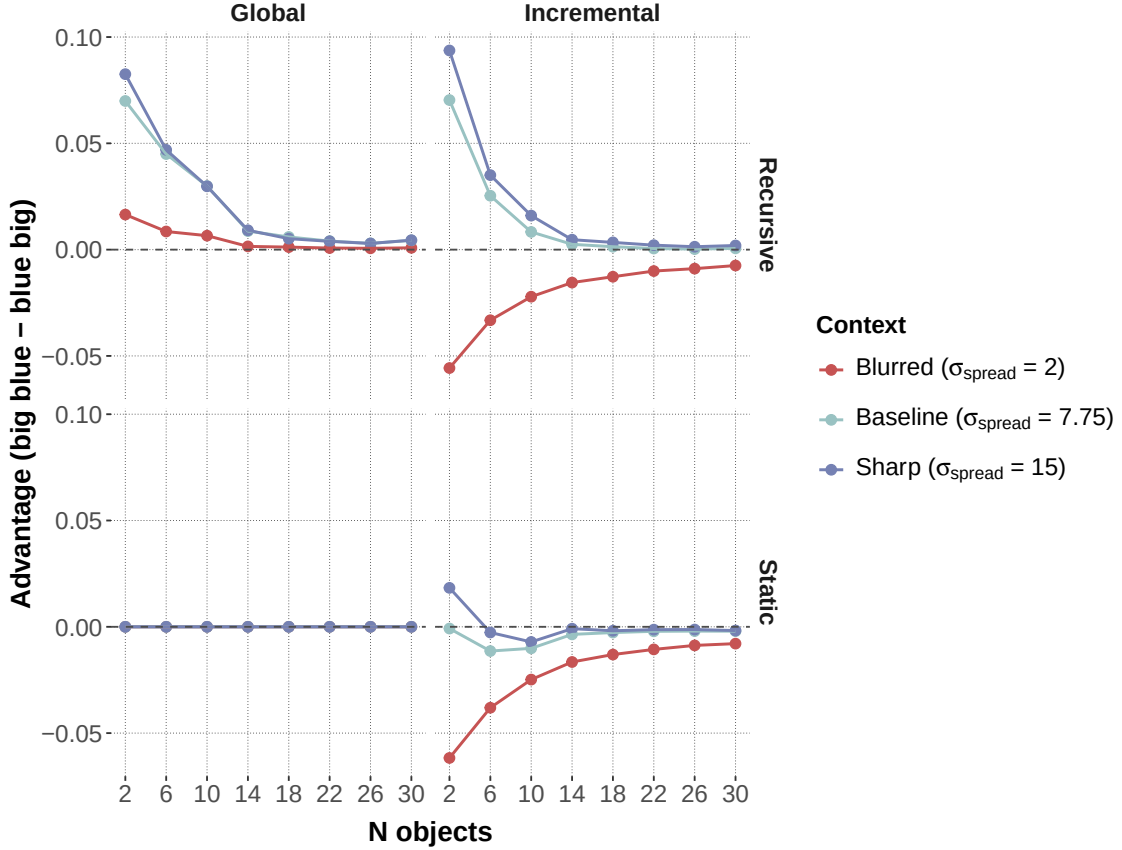
| Parameter                              | Values swept                 | # levels |
|----------------------------------------|------------------------------|----------|
| Speaker model                          | Incremental, global          | 2        |
| Semantics                              | Recursive, static            | 2        |
| $n_{\text{obj}}$                       | 2, 6, 10, 14, 18, 22, 26, 30 | 8        |
| $\sigma_{\text{spread}}$               | 2.0, 7.75, 15.0              | 3        |
| $\nu$                                  | 0.90, 0.92, 0.94, 0.96, 0.98 | 5        |
| $k$                                    | 0.10, 0.30, 0.50, 0.70, 0.90 | 5        |
| $w_f$                                  | 0.1, 0.2, 0.3, 0.5, 0.8      | 5        |
| Random context samples per combination | 1,000                        |          |
| <b>Total trials</b>                    | <b>12,000,000</b>            |          |

The parameter  $\sigma_{\text{spread}}$  controls the spread of the size distribution in the sampled context: low values produce low-discriminability contexts in which size is weakly informative, whereas high values produce high-discriminability contexts in which size strongly separates the target from its competitors. The remaining parameters ( $k$ ,  $w_f$ ,  $\nu$ ) are the fixed semantic parameters from the fitted models; their individual effects are reported in the Appendix.

## 7.2 Results

Figure 10 shows the size-first advantage as a function of context size and contextual spread, faceted by speaker model (columns) and semantic regime (rows).

Three findings stand out. First, under recursive semantics, both speaker models generate a size-first advantage across most of the parameter space. A global speaker with context-sensitive composition already derives the canonical ordering asymmetry without any incremental machinery. This result is cumulative with the subjectivity literature: once

**Figure 10**

*Size-first advantage as a function of context size ( $n_{obj}$ ) and contextual spread ( $\sigma_{spread}$ ), faceted by speaker model (columns) and semantic regime (rows).*

semantic values are evaluated over progressively restricted comparison classes, less noisy inner modification is sufficient to produce the familiar ordering direction (Franke et al., 2019; Scontras et al., 2020, 2019).

Second, static semantics completely eliminates the global speaker’s advantage. Under static semantics, the global speaker produced  $\Delta = 0.000$  across all parameter combinations without exception. When the size threshold does not update during composition, the literal listener assigns identical posteriors to both orderings, making them informationally equivalent. This reveals that the global speaker’s baseline advantage depends entirely on recursive semantics: it stems from order-sensitive threshold adaptation, not from the speaker

architecture itself.

Third, the incremental speaker amplifies the size-first advantage under recursive semantics in much of the parameter space and, crucially, generates qualitatively distinct predictions in a specific region: low-discriminability, low-cardinality contexts, where colour-first preferences emerge. Under static semantics, the incremental speaker defaults to colour-first ( $\Delta = -0.009$ ), driven by the chain-rule asymmetry that favours “blue” as the opening token when size is insufficiently diagnostic. This division of labour clarifies the architecture: context-sensitive composition provides the necessary asymmetry that the global speaker can exploit, while incremental production adds an online control policy whose ordering preference depends on the relative informativeness of each adjective at step 1.

## 8 General Discussion

Three findings characterise the results. First, the simulation shows that the global speaker yields a size-first advantage only under recursive semantics; under static semantics the advantage vanishes. Second, both incremental models outperform both global models in both datasets. Third, context-updating semantics improves fit in the production data but not in the slider data, and within the production data only in combination with an incremental speaker — a speaker  $\times$  semantics interaction rather than two independent effects.

The cross-task pattern is also informative. Context-updating composition helps the global models in the slider data, where it is the only available source of order sensitivity, but not in production, where the language-model prior already provides ordering information. Within the incremental models the pattern reverses: context-updating helps in production but not in slider ratings, where the binary response space does not discriminate between sharper and softer colour predicates. The benefit of recursive composition therefore appears to depend both on the absence of alternative order-relevant signals and on the richness of the alternative space.

The fitted colour reliability is  $\nu_{\text{colour}} \approx 0.59$ , a graded predicate rather than a near-deterministic one. Under a near-deterministic predicate the model has little room to

weight colour informationally, so unexplained variance accumulates in the uniform lapse term and the colour informativity weight collapses. Under the graded predicate the colour weight is substantially larger and the lapse rate substantially smaller.

The ablation models reported in the Appendix constrain the architecture further. The LM-only and lookahead variants are worse than the full incremental model, and the RSA-only variant fails PSIS-LOO diagnostics, indicating that both the lexical prior and the pragmatic component contribute to the predictive distribution over the 15-way response space.

Taken together, the framework brings subjectivity-based accounts of nominal modification, efficiency-based accounts of reference, and incremental pragmatic models into a single probabilistic model. Syntactic accounts that derive ordering from a fixed functional hierarchy (Cinque, 1994; Scott, 2002) capture the default size-before-colour pattern but not its graded, context-dependent departures. Subjectivity-based accounts (Franke et al., 2019; Scontras et al., 2017) capture the default but treat it as a stable property of adjective classes rather than a context-sensitive outcome. In the present framework the default ordering arises from the interaction between semantic noise and recursive composition, and context-dependent modulation from the incremental speaker’s sensitivity to the current discriminatory landscape.

## 9 References

- Cinque, G. (1994). On the evidence for partial N-movement in the Romance DP. In G. Cinque, J. Koster, J.-Y. Pollock, L. Rizzi, & R. Zanuttini (Eds.), *Paths towards universal grammar: Studies in honor of Richard S. Kayne* (pp. 85–110). Washington, DC: Georgetown University Press.
- CohnGordon, R., Goodman, N. D., & Potts, C. (2019). An incremental iterated response model of pragmatics. In *Proceedings of the society for computation in linguistics* (Vol. 2, pp. 81–90). doi: 10.7275/7yvv-jt08
- Degen, J. (2023). The rational speech act framework. *Annual Review of Linguistics*, 9, 519–540.
- Degen, J., Hawkins, R. D., Graf, C., Kreiss, E., & Goodman, N. D. (2020). When redundancy is useful: A bayesian approach to “overinformative” referring expressions. *Psychological Review*, 127(4), 591.
- Dixon, R. M. W. (1982). *Where have all the adjectives gone? and other essays in semantics and syntax*. Berlin: Mouton.
- Dyer, W., Torres, C., Scontras, G., & Futrell, R. (2023). Evaluating a century of progress on the cognitive science of adjective ordering. *Transactions of the Association for Computational Linguistics*, 11, 1185–1200.
- Feigenson, L., Dehaene, S., & Spelke, E. (2004). Core systems of number. *Trends in Cognitive Sciences*, 8(7), 307–314.
- Franke, M., Scontras, G., & Simoncic, M. (2019). Subjectivity-based adjective ordering maximizes communicative success. In *Cogsci* (pp. 344–350).
- Fukumura, K. (2018). Ordering adjectives in referential communication. *Journal of Memory and Language*, 101, 37–50.
- Futrell, R., Dyer, W., & Scontras, G. (2020, July). What determines the order of adjectives in English? comparing efficiency-based theories using dependency treebanks. In D. Jurafsky, J. Chai, N. Schluter, & J. Tetreault (Eds.), *Proceedings of the 58th annual*

- meeting of the association for computational linguistics* (pp. 2003–2012). Online: Association for Computational Linguistics. Retrieved from <https://aclanthology.org/2020.acl-main.181/> doi: 10.18653/v1/2020.acl-main.181
- Goodman, N. D., & Frank, M. C. (2016). Pragmatic language interpretation as probabilistic inference. *Trends in cognitive sciences*, 20(11), 818–829.
- Hahn, M., Degen, J., Goodman, N., Jurafsky, D., & Futrell, R. (2018). An information-theoretic explanation of adjective ordering preferences. In *Proceedings of the 40th annual meeting of the cognitive science society (cogsci)* (pp. 1766–1771). Retrieved from <https://cogsci.mindmodeling.org/2018/papers/0339/index.html>
- Hoffman, M. D., & Gelman, A. (2014). The no-u-turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15(47), 1593–1623.
- Martin, J. E. (1969). Semantic determinants of preferred adjective order. *Journal of Verbal Learning and Verbal Behavior*, 8(6), 697–704.
- Phan, D., Pradhan, N., & Jankowiak, M. (2019). Composable effects for flexible and accelerated probabilistic programming in numpyro. *arXiv preprint arXiv:1912.11554*.
- Quirk, R., Greenbaum, S., Leech, G., & Svartvik, J. (1972). *A grammar of contemporary english*. London: Longman.
- Rubio-Fernández, P. (2016). How redundant are redundant color adjectives? an efficiency-based analysis of color overspecification. *Frontiers in psychology*, 7, 153.
- Rubio-Fernandez, P. (2019). Overinformative speakers are cooperative: Revisiting the gricean maxim of quantity. *Cognitive science*, 43(11), e12797.
- Rubio-Fernandez, P., & Jara-Ettinger, J. (2020). Incrementality and efficiency shape pragmatics across languages. *Proceedings of the National Academy of Sciences*, 117(24), 13399–13404.



- Rubio-Fernández, P., Mollica, F., & Jara-Ettinger, J. (2021). Speakers and listeners exploit word order for communicative efficiency: A cross-linguistic investigation. *Journal of Experimental Psychology: General*, 150(3), 583–594.
- Schlotterbeck, F., & Wang, H. (2023). An incremental rsa model for adjective ordering preferences in referential visual context. *Proceedings of the Society for Computation in Linguistics*, 6(1), 121–132.
- Schmidt, L. A., Goodman, N. D., Barner, D., & Tenenbaum, J. B. (2009). How tall is tall? Compositionality, statistics, and gradable adjectives. In N. A. Taatgen & H. van Rijn (Eds.), *Proceedings of the 31st annual conference of the cognitive science society*.
- Scontras, G. (2023). Adjective ordering across languages. *Annual Review of Linguistics*, 9, 357–376.
- Scontras, G., Bar-Sever, G., Kachakeche, Z., Rosales Jr, C. M., & Samonte, S. (2020). Incremental semantic restriction and subjectivity-based adjective ordering. In *Proceedings of sinn und bedeutung* (Vol. 24, pp. 253–270).
- Scontras, G., Degen, J., & Goodman, N. D. (2017). Subjectivity predicts adjective ordering preferences. *Open Mind*, 1(1), 53–66.
- Scontras, G., Degen, J., & Goodman, N. D. (2019). On the grammatical source of adjective ordering preferences. *Semantics and Pragmatics*, 12, 7–1.
- Scott, G.-J. (2002). Stacked adjectival modification and the structure of nominal phrases. In G. Cinque (Ed.), *Functional structure in DP and IP: The cartography of syntactic structures, vol. 1* (pp. 91–120). Oxford: Oxford University Press.
- Sedivy, J. C., Tanenhaus, M. K., Chambers, C. G., & Carlson, G. N. (1999). Achieving incremental semantic interpretation through contextual representation. *Cognition*, 71(2), 109–147.
- Sproat, R., & Shih, C. (1991a). The cross-linguistic distribution of adjective ordering restrictions. In C. Georgopoulos & R. Ishihara (Eds.), *Interdisciplinary approaches to language: Essays in honor of S.-Y. Kuroda* (pp. 565–593). Kluwer.

- Sproat, R., & Shih, C. (1991b). The cross-linguistic distribution of adjective ordering restrictions. In *Interdisciplinary approaches to language: Essays in honor of s.-y. kuroda* (pp. 565–593). Springer.
- Sweet, H. (1898). *A new english grammar, logical and historical, vol. 2: Syntax*. London: Clarendon.
- Tanenhaus, M. K., Spivey-Knowlton, M. J., Eberhard, K. M., & Sedivy, J. C. (1995). Integration of visual and linguistic information in spoken language comprehension. *Science*, 268(5217), 1632–1634.
- Teodorescu, A. (2006). Adjective ordering restrictions revisited. In D. Baumer, D. Montero, & M. Scanlon (Eds.), *Proceedings of the 25th West Coast Conference on Formal Linguistics* (pp. 399–407). Somerville, MA: Cascadilla Proceedings Project.
- Trainin, N., & Shetreet, E. (2021). It’s a dotted blue big star: on adjective ordering in a post-nominal language. *Language, Cognition and Neuroscience*, 36(3), 320–341.
- Trotzke, A., & Wittenberg, E. (2019). Long-standing issues in adjective order and corpus evidence for a multifactorial approach. *Linguistics*, 57(2), 273–282. doi: 10.1515/ling-2019-0001
- Vehtari, A., Gelman, A., & Gabry, J. (2017). Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and Computing*, 27(5), 1413–1432.
- Waldon, B., & Degen, J. (2021). Modeling cross-linguistic production of referring expressions. In *Proceedings of the society for computation in linguistics 2021* (pp. 206–215). Association for Computational Linguistics.
- Whorf, B. L. (1945). Grammatical categories. *Language*, 21(1), 1–11.

## Appendix A

### Production Ablation Models

In addition to the parameter-matched  $2 \times 2$  reported in the Model Comparison section, we fitted three ablation variants of the core incremental speaker (single  $\alpha$  plus language-model prior, before the extensions of Section 6.2) to the production data in order to isolate the contributions of specific architectural components. All ablation models use hierarchical participant pooling.

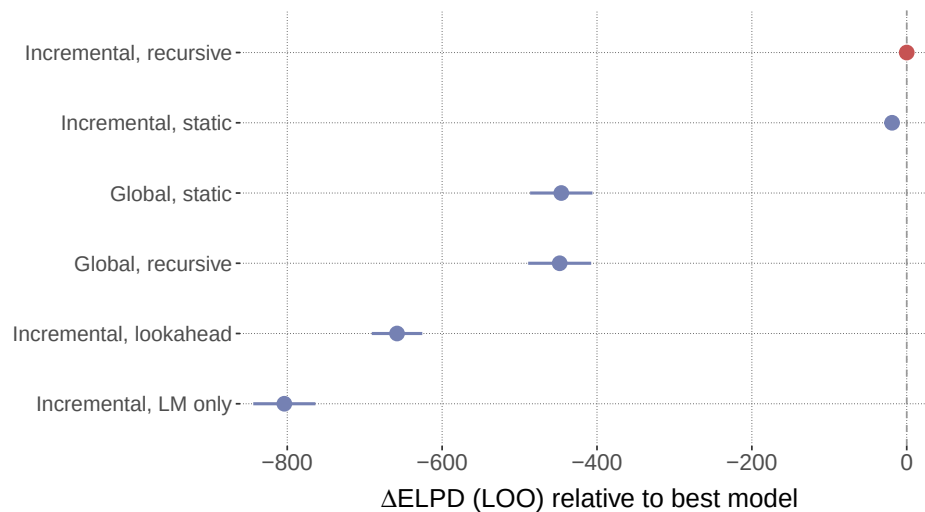
**Incremental lookahead.** Instead of choosing greedily at each word position, this variant evaluates each possible continuation by looking one step ahead and selecting the word that maximizes expected utility over the next two positions. It tests whether myopic word-by-word choice is sufficient or whether a short planning horizon improves fit.

**LM only.** This variant sets the RSA rationality parameter  $\alpha = 0$ , removing all pragmatic reasoning. Utterance choice is driven entirely by the language-model prior over word sequences, testing whether phrase-frequency information alone can account for ordering preferences.

**RSA only.** This variant sets the language-model weight  $\beta = 0$ , removing the utterance prior entirely. All ordering information must come from the RSA informativeness term and participant-level pragmatic strength. This model was successfully fit, but PSIS-LOO produced non-finite results because a substantial subset of observations have very high Pareto- $k$  values ( $k > 0.7$  for 199 of 3,196 observations). This indicates that the leave-one-out posterior is too different from the full posterior for reliable importance-sampling reweighting. The RSA-only model is therefore excluded from the ranked comparison but is noted here for completeness.

Figure A1 shows the  $\Delta\text{ELPD}$  comparison for the four main models and the two ablation variants with usable PSIS-LOO estimates. The incremental context-updating model remains clearly best. The lookahead variant performs worse than both global models, suggesting that greedy one-step choice is more appropriate than short-horizon planning for

this task. The LM-only variant is the worst-fitting model overall, confirming that ordering preferences cannot be reduced to conventional phrase-frequency effects.



**Figure A1**

*Production ablation model comparison.  $\Delta\text{ELPD}$  relative to the best model (incremental, context-updating) with difference standard errors; RSA-only ablation excluded (non-finite PSIS-LOO).*

## Appendix B

### Extended Production Model: Parameter Inventory

Table B1 lists the ten population parameters of the advocated production model (Section 6.2), their priors, and their posteriors at the reference ten-parameter fit. The participant hierarchy adds 339 latent offsets ( $113 \text{ participants} \times 3 \text{ conditions}$ ) drawn from a zero-mean normal with standard deviation  $\tau$ . The fixed constants are  $\nu_{\text{colour}} = 0.59$ ,  $\nu_{\text{form}} = 0.50$ ,  $k = 0.50$ ,  $w_f = 0.69$ , and the language-model temperature fixed at its stable posterior median. The posteriors below are from the reference fit ( $\nu_{\text{colour}} = 0.97$ ), at which every coefficient is documented; under the calibrated  $\nu_{\text{colour}} = 0.59$  the four most colour-sensitive coefficients shift to  $\alpha_S \approx 8.0$ ,  $\alpha_C \approx 3.8$ ,  $\lambda_{\text{noncanon}} \approx 4.4$ , and  $\epsilon \approx 0.11$ , and every 94% HDI continues to exclude zero.

**Table B1**

*Population parameters of the advocated production model: prior and posterior mean with 94% HDI at the reference ten-parameter fit.*

| Parameter                   | Role                                          | Prior           | Posterior            |
|-----------------------------|-----------------------------------------------|-----------------|----------------------|
| $\alpha_S$                  | RSA informativity weight, size                | HalfNormal(5)   | 6.24 [6.02, 6.48]    |
| $\alpha_C$                  | RSA informativity weight, colour              | HalfNormal(5)   | 1.26 [0.95, 1.58]    |
| $\lambda_{\text{suff}}$     | first-word boost for the sufficient dimension | Normal(0, 1)    | 1.58 [1.19, 1.97]    |
| $\lambda_{\text{form-mod}}$ | context-gated form-mention boost              | Normal(0, 2)    | 2.82 [2.62, 3.05]    |
| $\lambda_{\text{noncanon}}$ | form-before-colour penalty                    | HalfNormal(2)   | 2.82 [2.21, 3.46]    |
| $\gamma_{\text{base}}$      | base per-word length bias                     | Normal(0, 2)    | 3.10 [2.94, 3.26]    |
| $\gamma_{\text{one}}$       | length modulation when one word suffices      | Normal(0, 2)    | -2.37 [-2.57, -2.19] |
| $\gamma_{\text{sharp}}$     | extra length under perceptual blur            | HalfNormal(2)   | 0.68 [0.53, 0.82]    |
| $\epsilon$                  | lapse / uniform-noise rate                    | Beta(1, 50)     | 0.18 [0.17, 0.20]    |
| $\tau$                      | SD of participant rationality offsets         | HalfNormal(0.2) | 1.35 [1.12, 1.59]    |

## Appendix C

### Production Model Selection: Parsimony Frontier and Constant Calibration

**A parsimony frontier.** The component set was not chosen by hand. We fitted the maximal thirteen-parameter model and walked a leave-one-out frontier, dropping coefficients that carry no fit information and stopping at the first removal that did cost fit (Table C1). The frontier floor is a ten-parameter model: every retained coefficient has a 94% HDI excluding zero, and removing any of the remaining ten —  $\lambda_{\text{suff}}$  and  $\gamma_{\text{sharp}}$  in particular — measurably degrades fit. We report this ten-parameter model, with the maximal model and the frontier itself as the selection evidence.

**Table C1**

*Parsimony frontier for the extended production speaker (size-colour subset, four chains, hierarchical).  $R_{\text{all}}^2$  uses all condition  $\times$  utterance cells;  $R_{\geq .02}^2$  restricts to behaviourally relevant cells (empirical proportion  $\geq 0.02$ );  $r$  is the Pearson correlation between predicted and observed proportions. The first three rows remove coefficients at zero fit cost; the last two show the frontier floor, where further removal costs fit.*

| Model                                 | #named    | $R_{\text{all}}^2$ | $R_{\geq .02}^2$ | $r$          |
|---------------------------------------|-----------|--------------------|------------------|--------------|
| Maximal model                         | 13        | 0.919              | 0.919            | 0.959        |
| – LM temperature                      | 12        | 0.920              | 0.919            | 0.959        |
| – LM temp. – 3-word penalty           | 11        | 0.920              | 0.920            | 0.959        |
| – also $\alpha_F$ ( <b>reported</b> ) | <b>10</b> | <b>0.921</b>       | <b>0.921</b>     | <b>0.959</b> |
| – also $\lambda_{\text{suff}}$        | 10        | 0.909              | 0.904            | 0.954        |
| – also $\gamma_{\text{sharp}}$        | 10        | 0.900              | 0.889            | 0.949        |

**Calibrating the fixed semantic constants.** The ten-parameter model still depends on four constants held fixed during inference: the colour and form reliabilities  $\nu_{\text{colour}}$ ,  $\nu_{\text{form}}$ , the size-threshold anchor  $k$ , and the size-sigmoid sharpness  $w_f$ . We audited all four jointly (Table C2). A static sweep over one constant is misleading because the constant trades off against the fitted parameters; only a joint refit licenses a conclusion about its

value. Under joint refit, only  $\nu_{\text{colour}}$  warranted a changed value, moving from the originally assumed 0.97 to 0.59 [0.56, 0.62]; re-fixing it at 0.59 recovers the gain at ten parameters with clean sampling. The other three constants did not move identifiably, and freeing all four jointly produced a non-identified ridge, which we take as a caution against over-parameterising the semantics.

**Table C2**

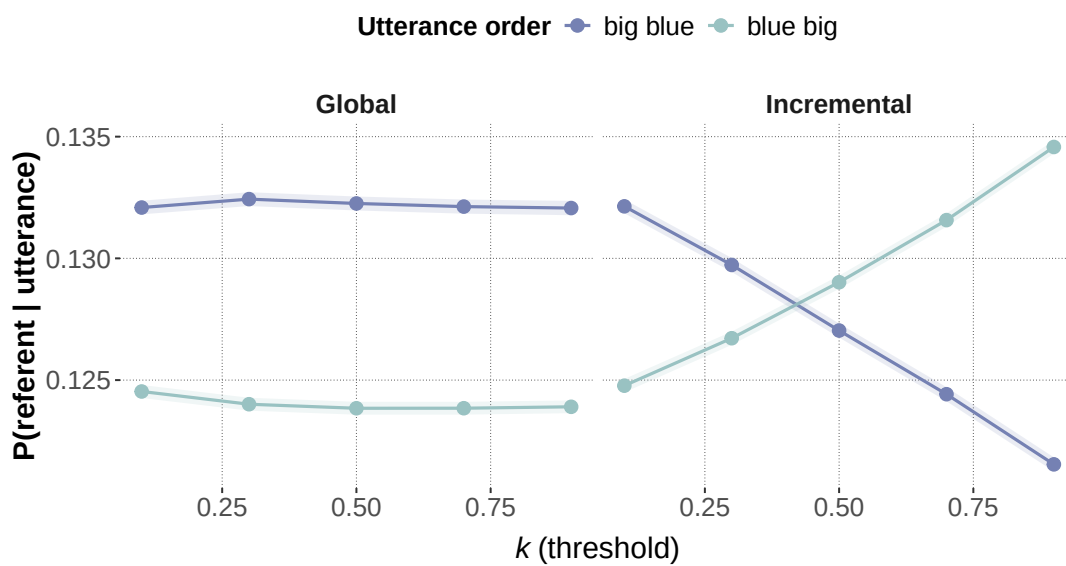
*Fixed-constant sensitivity analysis for the ten-parameter production model (joint refits, size-colour subset). Each “free” row frees one constant and refits all parameters jointly; “free all four” frees them together.  $\Delta R^2$  is relative to the reported model. Only  $\nu_{\text{colour}}$  warrants a changed value; it is re-fixed at its freed posterior mean.*

| Refit                                                               | #named    | $R^2_{\text{all}}$ | Freed posterior / verdict       |
|---------------------------------------------------------------------|-----------|--------------------|---------------------------------|
| Reported ( $\nu_{\text{colour}} = 0.97$ , all fixed)                | 10        | 0.921              | —                               |
| Free $\nu_{\text{colour}}$                                          | 11        | 0.942              | 0.59 [0.56, 0.62]; large gain   |
| Free $\nu_{\text{form}}$                                            | 11        | 0.920              | unidentified (HDI spans prior)  |
| Free $k$                                                            | 11        | 0.921              | negligible gain                 |
| Free $w_f$                                                          | 11        | 0.921              | brackets fixed value            |
| Free all four                                                       | 14        | 0.953              | non-identified ridge (ESS 6–28) |
| <b><math>\nu_{\text{colour}}</math> re-fixed at 0.59 (reported)</b> | <b>10</b> | <b>0.942</b>       | clean sampling, $r = 0.97$      |

## Appendix D

### Simulation Parameter Sweeps

Figures D1 to D3 show how the three fixed semantic parameters affect communicative success across the two speaker models.



**Figure D1**

*Effect of the threshold parameter  $k$  on communicative success in the simulation.*



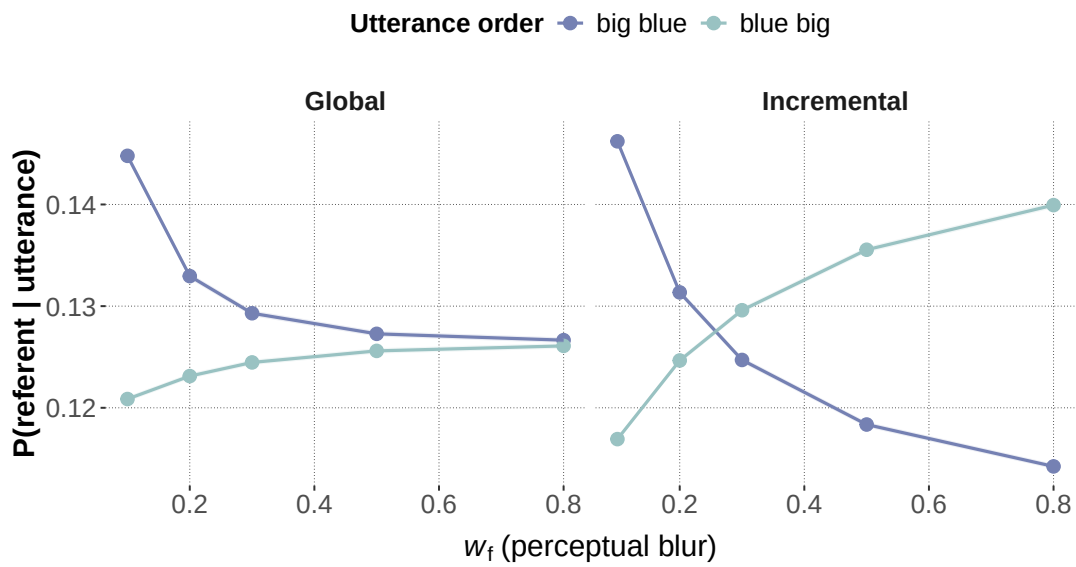


Figure D2

*Effect of perceptual blur  $w_f$  on communicative success in the simulation.*

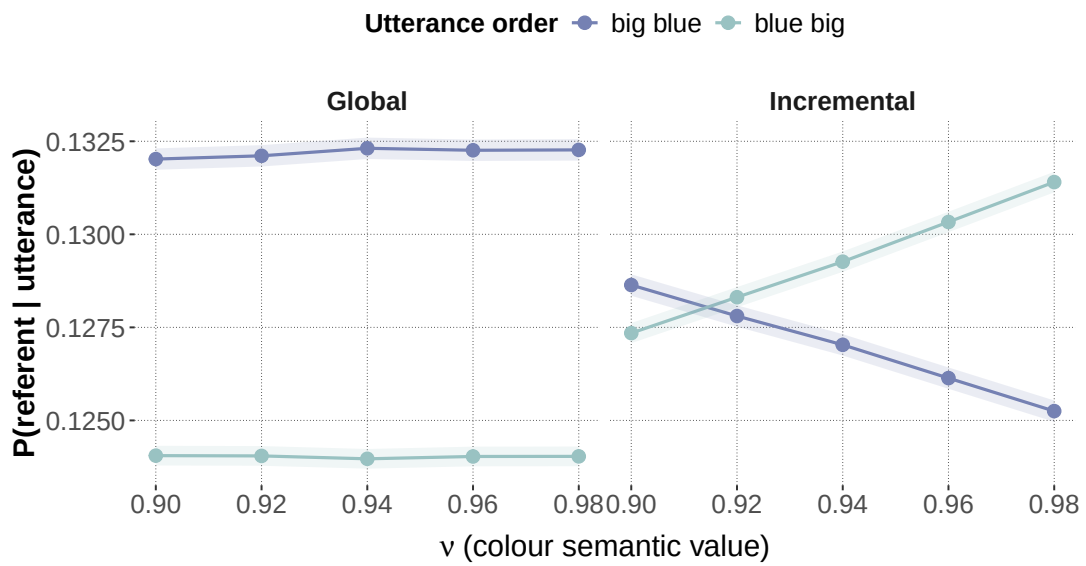


Figure D3

*Effect of the colour/form semantic value  $v$  on communicative success in the simulation.*

## Appendix E

### Slider Colour-Reliability Audit

The slider-rating model fixes the colour and form reliability  $\nu$  at 0.80 (Section 6).

Production already imposes a calibration step that lets the data choose  $\nu_{\text{colour}}$  (Section 6.2); for parity we apply the same audit to the slider’s incremental models. The audit samples  $\nu \sim \text{Uniform}(0.5, 0.999)$  within the model and recomputes the literal listeners at each MCMC step, while keeping every other component of the incremental hierarchical speaker unchanged. Table E1 reports the resulting posteriors alongside the originally fixed- $\nu$  fits.

**Table E1**

*Slider colour-reliability audit. Fixed- $\nu$  rows use  $\nu = 0.80$ ; free- $\nu$  rows sample  $\nu \sim \text{Uniform}(0.5, 0.999)$  and recompute literal listeners each step. “ELPD” is PSIS-LOO;  $\Delta\text{ELPD}$  and dse are relative to the incremental, context-updating fixed- $\nu$  model. All four models use 4 chains  $\times$  500 warmup  $\times$  500 samples; maximum split- $\hat{R} \leq 1.01$  for the free- $\nu$  models.*

| Cell                                                        | ELPD            | $\Delta\text{ELPD}$ | dse  | $\alpha$ (sd)      | $b$ (sd)           | $\nu$ (94% HDI)   |
|-------------------------------------------------------------|-----------------|---------------------|------|--------------------|--------------------|-------------------|
| <b>Incremental, updating, fixed <math>\nu = 0.80</math></b> | <b>-2,774.1</b> | <b>0.0</b>          | —    | <b>0.69 (0.06)</b> | <b>0.58 (0.06)</b> | —                 |
| Incremental, fixed, fixed $\nu = 0.80$                      | -2,774.1        | -0.06               | 0.18 | 0.69 (0.06)        | 0.58 (0.06)        | —                 |
| Incremental, updating, free $\nu$                           | -2,775.0        | -0.89               | 1.08 | 1.53 (0.61)        | 0.41 (0.15)        | 0.63 [0.50, 0.82] |
| Incremental, fixed, free $\nu$                              | -2,774.3        | -0.23               | 1.00 | 1.48 (0.55)        | 0.42 (0.13)        | 0.63 [0.51, 0.79] |

Three points stand out. First,  $\nu$  is weakly identified in the slider data: the posterior bumps the prior lower wall at 0.5 and only just excludes the fixed value at the upper end, in contrast to the tight production posterior 0.59 [0.56, 0.62]. Second, the four models are statistically indistinguishable in PSIS-LOO ( $|\Delta\text{ELPD}|/\text{dse} < 1$  throughout), so out-of-sample fit cannot decide between sharper and softer colour predicates in this response

space. Third, under the freed posterior the rationality parameter  $\alpha$  moves to roughly 1.5 (from 0.69) and the bias intercept  $b$  moves below 0.5 (from 0.58), exactly because  $\nu$  and the other parameters trade off: a softer colour predicate forces a stronger rationality term and reduces the residual size-first bias absorbed by  $b$ . We therefore retain the fixed  $\nu = 0.80$  for the slider model in the main text and treat  $\alpha$  and  $b$  at that specification as the reported quantities, while flagging the conditional status of the  $b > 0.5$  finding (Section 6, Slider-Rating Study). The cross-dataset asymmetry — production strongly identifies  $\nu$  at a value with substantive parameter consequences, the slider does not — is itself diagnostic of the response-space sensitivity argued for in the Cross-Dataset Summary.